Structural Transformation as Primitive Accumulation

A Thin Description of Subsistence, Entitlement, and Agricultural Commercialization in the United States

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The New School



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Question, Framework, and Evidence

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A first example: output and access can move differently

Intuition

Consider a household that works rented farmland and keeps part of the harvest.

- A new operator uses machinery to produce more output with fewer workers.
- The original household may obtain a better-paid job elsewhere, or it may lose a reliable source of food and income.
- Both paths can produce the same aggregate observation: less agricultural employment and more output per worker.

The paper asks what additional information helps interpret that aggregate change.

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Five terms used throughout the lecture

Productivity	Output obtained from measured inputs. Labor productivity uses labor alone in the denominator.
Structural transformation	Changes in the allocation of labor and production across sectors.
Provisioning	The arrangements through which people obtain the necessities of life.
Entitlement	Effective command over food or the resources needed to obtain it.
Primitive accumulation	The historical separation of producers from productive resources and the reorganization of their dependence on markets.

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The question of agricultural transformation

Wang (2026)

- Rising output per worker allows agriculture to produce with less labor.
- The same transition can change who controls land and how households obtain food.
- The paper asks whether historical commercialization is associated with weaker agriculture-centered provisioning and greater contemporary reliance on public food assistance.
- Productive performance and command over a livelihood are related but different objects of inquiry.

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The full presentation

1. The question, historical motivation, and thin description.
2. **Slice 1: Productivity and entitlement.** Structural calibration, historical tenure, and agricultural labor absorption.
3. **Slice 2: Profit and subsistence.** Farm dynamics, the relative-margin index, and the growth comparison.
4. **Slice 3: Cash crops and SNAP.** Commercialization, state estimates, and the county extension.
5. Interpretation, sensitivity checks, and worked examples.

▸ Accounting ▸ Slice 1 checks ▸ Slice 2 checks ▸ Slice 3 checks
▸ Connections ▸ References

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Three meanings of subsistence

Biological necessities	Food and other goods required to sustain human life.
Formal model parameter	A positive agricultural consumption term $\bar{a}_s$ in Stone–Geary preferences.
Historical interpretation	The relations through which households obtain necessities and maintain agricultural livelihoods.

The calibrated parameter provides a model-dependent index of agriculture-centered subsistence. It does not directly measure hunger or entitlement security.

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The paper's argument in three connected steps

1. **Access and subsistence.** The calibrated subsistence index is positively associated with tenant-oriented historical tenure composition.
2. **Control and surplus.** In the closed-resource model, expansion transfers an existing operating position. The fitted margin index is associated with lower subsistence and faster later labor-productivity growth.
3. **Commercialization and assistance.** The significant baseline soybean IV relationship overlaps with historical tenure; further adjustment for fieldwork conditions mainly reduces precision.

Primitive accumulation concerns whose capacity to produce, work, and retain returns changes as agriculture becomes more productive.

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Sen: command over food

Sen (1981)

- Food availability describes the supply of food.
- Entitlements concern the food a person can command through resources and socially recognized claims.
- Production, paid work, exchange, and public transfers can all support that command.
- A rise in aggregate output alone does not establish whether a particular household has gained secure access.

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Marx: separation from productive resources

Marx (1976, Part VIII)

- Primitive accumulation directs attention to the historical separation of producers from the means of production and subsistence.
- Market participation can expand when independent means of provisioning contract.
- Output growth can accompany increasing dependence on wages and commodity exchange.
- The paper applies this question to U.S. institutions. It does not equate U.S. commercialization with a literal repetition of English enclosure.

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Thin description as an analytical orientation

Jackson (2013)

- Jackson approaches subjects as interpreters of their own circumstances, not simply objects for an exhaustive outside account.
- I adapt this orientation by connecting partial quantitative accounts with historical testimony.
- Each slice makes a particular relationship visible. Greater detail within it does not resolve every question outside its frame.
- Together, the slices ask how production changes, who retains operating control, and how households secure subsistence.

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Three thin slices of agricultural transformation

Paper location	Evidence and method	Question
1. Productivity and entitlement Paper Section 3	Labor and productivity, 1960–2010; tenure, 1940. Two-sector calibration and tenure comparisons.	How does agriculture absorb labor, and how does that dependence relate to tenure?
2. Profit and subsistence Paper Section 4	Farm-size distributions, 1940 and 1950. Distributional calibration and conditional margins.	How does operating structure relate to provisioning and subsequent growth?
3. Cash crops and SNAP Paper Section 5	Historical crops and contemporary SNAP. State OLS and 2SLS, with a county reduced form.	How does commercialization relate to public food assistance?

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Two accounts in dialogue

Efficiency and growth	Primitive accumulation
Productivity gains and specialization support labor reallocation and growth.	Changing access to productive resources reorganizes dependence on wages, markets, and public transfers.
The organizing question concerns productive performance and allocation.	The organizing question concerns operating control and command over a livelihood.

Shared question: How do productive gains intersect with changing claims to subsistence?

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Reading the equations without prior training

s, t	A state and a time period. A subscript identifies which observation a symbol describes.
$\sum_s x_s$	Add the values of x across states. Dividing by their number gives a mean.
$\widehat{x}$	A fitted or estimated value, distinguished from an observed quantity.
$\log x$	The natural logarithm. Log differences describe proportional changes.
$\partial y / \partial x$	The local change in y when x changes and specified other quantities stay fixed.
$\mathbb{E}[x]$	An expected value: an average over possible outcomes under a probability model.

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Three kinds of claims in this lecture

Accounting identity	True by definitions and algebra, such as revenue minus costs. It does not identify a causal mechanism.
Model implication	Follows from stated assumptions, such as a labor-share equation or a conditional margin ranking.
Empirical association	Describes how observed or constructed variables covary in the sample. A causal interpretation requires further assumptions.

The historical interpretation connects these different kinds of evidence without treating them as interchangeable.

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Dates correspond to different empirical objects

1919	County soybean acres, reported in historical agricultural tabulations.
1926–1940	Average state crop acreage for soybean, corn, and cotton exposure.
1940	Farm-operator tenure composition and the entry-profile calibration.
1940 and 1950	Separate farm-size distributions for fitting p and constructing $\mathcal{M}$.
1960 and 1970	Baseline early-year recovery of $\bar{a}_s(\alpha)$.
1960–2010	Six decennial labor-share and productivity observations per state.
1995–2026	Mean fieldwork suitability across available April–October weeks.
2024 five-year ACS	County household SNAP receipt and controls.
February 2026	State SNAP participants, divided by July 2025 population.

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Why these historical windows serve different purposes

- **1940 tenure** describes inherited forms of agricultural access. **1940 and 1950 farm counts** discipline the historical distribution of operating control.
- **1960 and 1970** supply early subsistence-recovery moments. The full **1960–2010** panel selects the common preference parameter and supports trajectory comparisons.
- **1926–1940 state soybeans** and **1919 county soybeans** measure different historical commercialization exposures.
- Contemporary SNAP measures later public provisioning. It does not follow individual displaced producers into assistance.

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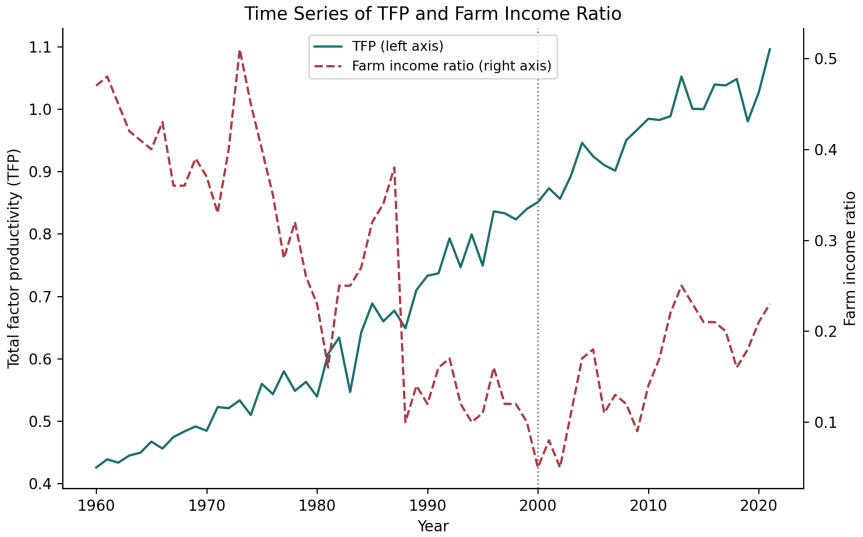
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§2
Productivity, Income, and Historical Change

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Productivity and farm household income



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The farm income share has a household denominator

$$\text{FarmShare}_t = \frac{\text{average farm income}_t}{\text{total average farm-household income}_t}.$$

- The ratio concerns the composition of income among farm households.
- Off-farm earnings affect the denominator.
- It differs from labor's share of agricultural value added or farms' share of national income.
- A change in which households remain farm households can affect its interpretation.

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How national TFP growth is measured

Wang et al. (2024)

The Törnqvist index uses adjacent-period average value shares:

$$\Delta \log TFP_t = \sum_i \frac{R_{it} + R_{i,t-1}}{2} \Delta \log Y_{it} - \sum_j \frac{W_{jt} + W_{j,t-1}}{2} \Delta \log X_{jt}.$$

$$R_{it} = \frac{P_{it} Y_{it}}{\sum_h P_{ht} Y_{ht}}, \qquad W_{jt} = \frac{v_{jt} X_{jt}}{\sum_k v_{kt} X_{kt}}.$$

- Output growth enters with revenue shares and input growth with cost shares.
- Quality adjustments enter the input quantities where available.
- National TFP is distinct from the state labor-productivity ratio used in calibration.

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Labor productivity and TFP answer different questions

Intuition

- Labor productivity asks how much output each unit of labor produces.
- Replacing workers with more machinery can raise labor productivity even if the efficiency of the entire input bundle changes little.
- TFP compares output with a measured bundle that also includes nonlabor inputs.
- The motivation uses national TFP. The structural calibration uses state agricultural labor productivity.

$$A_a = \frac{\text{output quantity index}}{\text{labor input quantity index}}$$ is not the national TFP index.

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The revenue–cost accounting connection

Let $V = \sum_i P_i Y_i$, $C = \sum_j v_j X_j$, and $H = V/C$.

$$d\log V = \sum_i R_i (d\log P_i + d\log Y_i),$$

$$d\log C = \sum_j W_j (d\log v_j + d\log X_j).$$

Subtracting and rearranging gives the continuous-time identity

$$\underbrace{\sum_i R_i \, d\log Y_i - \sum_j W_j \, d\log X_j}_{\text{Divisia TFP growth}} = d\log H + \sum_j W_j \, d\log v_j - \sum_i R_i \, d\log P_i.$$

▸ Worked accounting

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Productivity accounting and surplus claims

Shaikh (1974)

- At unchanged prices, more output from the same measured inputs raises TFP and V/C.
- The surplus above measured costs as a share of revenue is $1 - C/V = 1 - H^{-1}$.
- Some land and financial returns already enter input costs. The residual depends on the accounting boundary.
- Income redistribution alone, with quantities unchanged, does not raise quantity-based TFP.
- Shaikh's critique motivates care in interpreting an accounting residual as independently identified technical progress.

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A descriptive break around 2000

$$TFP_t = b_0 + b_1 \mathrm{FarmShare}_t + b_2 D_t + b_3 (\mathrm{FarmShare}_t D_t) + \varepsilon_t, \quad D_t = \mathbf{1}\{t \geq 2000\}.$$

- Before 2000, the fitted slope is b_1.
- From 2000, the fitted slope is $b_1 + b_3$.
- The dependent variable is the TFP index level, not its annual growth rate.
- The break regression summarizes time-series association. It is not a causal estimate of productivity's effect on household income.

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The reported break estimates

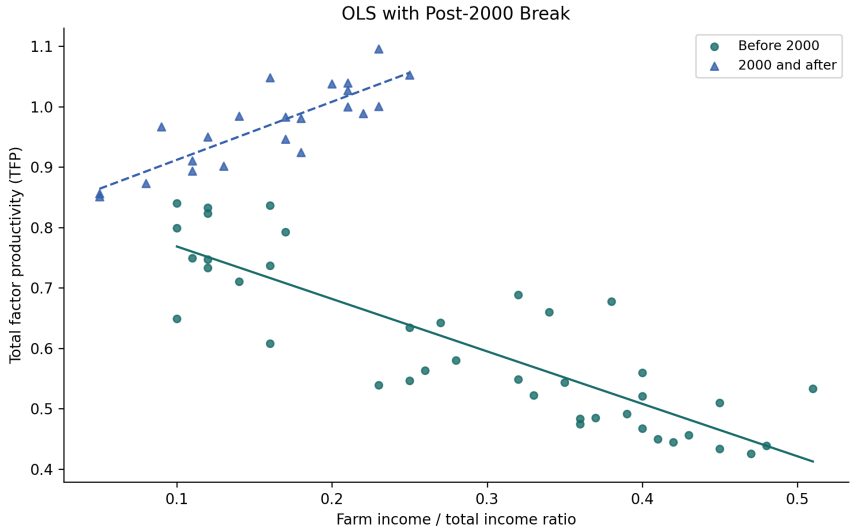
	Baseline OLS	Break specification
Farm income share	-1.206^{***}	-0.868^{***}
	(0.147)	(0.075)
Post-2000 indicator		-0.039
		(0.044)
Interaction		1.829^{***}
		(0.233)
Implied post-2000 slope		0.961^{***}
		(0.220)
R^2	0.529	0.921
Years	62	62

Standard errors in parentheses; $^{***}p < 0.01$.

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The break pattern in the paper



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Family farming as a statistical category

- A family farm can be large, capital-intensive, and commercially specialized.
- Family ownership of the business need not imply control over contracts, processing, credit, or marketing.
- Investors can own land while a separate family business operates it.
- Farm counts, acreage, legal ownership, and control over returns describe different distributions.

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Historical shocks and differentiated mechanisms

1980s farm crisis	Debt-financed expansion met higher interest rates, weaker exports, and falling land values. Distress could force land sales and exits.
2007–08 crises	Commodity-price changes and investors' search for tangible assets increased interest in farmland income and appreciation.
Ownership and leasing	Investment ownership can separate claims to land rents and appreciation from the business cultivating the land.

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Why soybean specialization receives attention

- Soybeans fit existing grain-farming machinery and commercial Corn Belt production.
- Feed, oil, and industrial demand connected production to wider commodity markets.
- The paper treats historical soybean exposure as a marker of a commercial regime.
- Corn and cotton provide comparisons with different production histories.
- Crop-specific technology is a competing or complementary explanation, not something the calibration eliminates.

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Economic detail still requires interpretive choices

- Misallocation research asks how differences in resource allocation contribute to productivity and development.
- The choice of production unit, accounting representation, and selection mechanism helps define the question that detailed data answer.
- A family-farm classification concerns business ownership. It does not, by itself, describe scale, household labor, contractual autonomy, or livelihood security.
- Thin description connects these distinct questions instead of treating one representation as exhaustive.

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Slice 1: representation and welfare

- Herrendorf et al. (2013) find different roles for income and price effects in final-expenditure and value-added representations of U.S. structural change.
- Input–output relationships reconcile those representations. More detailed estimation does not eliminate the need to specify what the measured object represents.
- Baqaee and Burstein (2023) show how nonhomothetic preferences and changing tastes can separate measured growth from welfare improvement.
- I use calibrated subsistence to ask a related question about agricultural livelihoods, rather than interpreting productivity performance as a complete welfare measure.

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Slice 2: production units and selection

- Gollin and Udry (2021) show that measurement error and unobserved heterogeneity account for substantial measured agricultural productivity dispersion.
- Aragón et al. (2024) find larger misallocation estimates using plots than farms. A farm can share management and fixed resources across plots.
- Decreasing returns within existing units do not settle aggregate returns when entry, replication, and organization change.
- Foster et al. (2008) distinguish demand-driven selection from physical efficiency in manufacturing. This motivates asking what farm survival alone can establish.

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Slice 3: participation and socioeconomic conditioning

- Finkelstein and Notowidigdo (2019) show that information and application assistance affect SNAP enrollment in their experiment.
- Participation therefore depends on access and take-up as well as material need.
- Cinelli et al. (2024) explain how controlling for intermediate outcomes can block pathways within a relationship of interest.
- Income, employment, and access to assistance enter the provisioning relationship itself. The county specifications show how the association changes when these dimensions enter the conditioning set.

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Productivity and Entitlement

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Two sectors and one labor resource

$$Y_{a,s,t} = A_{a,s,t} L_{a,s,t}, \qquad Y_{n,s,t} = A_{n,s,t} L_{n,s,t},$$
$$L_{a,s,t} + L_{n,s,t} = L_{s,t}.$$

- Agriculture and non-agriculture produce with labor linearly.
- Each person supplies one unit of labor. Consumption is per person.
- The non-agricultural price equals one. The agricultural relative price is p_a.
- Both sectors operate in an interior competitive equilibrium.

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The two-sector equilibrium in ordinary language

Intuition

1. Households decide how much agricultural and non-agricultural output to consume.
2. Firms hire labor when its value in production covers the wage.
3. A common wage and relative prices make both active sectors willing to employ workers.
4. The amounts produced must equal the amounts households consume.

The agricultural labor share must satisfy all these conditions together. It is not an equation imposed independently on household demand.

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Stone–Geary preferences

(Kongsamut et al. 2001; Herrendorf et al. 2014)

$$U(c_a, c_n) = \log(c_a - \bar{a}) + \psi \log c_n, \qquad \bar{a} > 0, \quad \psi > 0.$$

The household chooses $c_a > \bar{a}$ and $c_n > 0$ subject to

$$p_a c_a + c_n = w.$$

- The shift $\bar{a}$ makes demand non-homothetic.
- The parameter describes a fixed agricultural consumption component inside the model.
- The empirical interpretation concerns agriculture-centered provisioning rather than a literal biological requirement.

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Household demand: the first-order conditions

$$\mathcal{L} = \log(c_a - \bar{a}) + \psi \log c_n + \lambda(w - p_a c_a - c_n).$$

$$\frac{1}{c_a - \bar{a}} = \lambda p_a, \qquad \frac{\psi}{c_n} = \lambda.$$

Dividing the conditions gives

$$c_n = \psi p_a (c_a - \bar{a}).$$

Substitution into the budget yields

$$p_a(1 + \psi)(c_a - \bar{a}) = w - p_a \bar{a}.$$

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Supernumerary income and demand

Define $\alpha = 1/(1+\psi) \in (0,1)$ and supernumerary income $w - p_a\bar{a}$.

$$c_a = \bar{a} + \alpha\left(\frac{w}{p_a} - \bar{a}\right), \qquad c_n = (1-\alpha)(w - p_a\bar{a}).$$

- α is the agricultural expenditure share of income remaining after the fixed agricultural component.
- It is not the total food expenditure share at every income level.
- Positivity requires $w > p_a\bar{a}$.

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Firm optimization and relative prices

Firms solve the linear profit problems

$$\max_{L_a \geq 0} (p_a A_a - w) L_a, \qquad \max_{L_n \geq 0} (A_n - w) L_n.$$

Positive finite production in both sectors requires zero unit profits:

$$p_a A_a = w = A_n, \qquad p_a = \frac{A_n}{A_a}, \qquad \frac{w}{p_a} = A_a.$$

The cancellation of A_n in the labor-share equation is an equilibrium algebra result. It does not mean non-agriculture is irrelevant to livelihoods.

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Deriving agricultural labor absorption

Agricultural goods-market clearing gives $c_a L = Y_a = A_a L_a$.

$$\ell_a \equiv \frac{L_a}{L} = \frac{c_a}{A_a}.$$

Using household demand and $w/p_a = A_a$,

$$c_a = \bar{a} + \alpha(A_a - \bar{a}) = \alpha A_a + (1-\alpha)\bar{a}.$$

Therefore

$$\ell_{a,s,t} = \alpha + (1-\alpha)\frac{\bar{a}_s}{A_{a,s,t}}.$$

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Comparative statics of the labor share

$$\frac{\partial \ell_a}{\partial A_a} = -\frac{(1-\alpha)\bar{a}}{A_a^2} < 0, \qquad \frac{\partial \ell_a}{\partial \bar{a}} = \frac{1-\alpha}{A_a} > 0.$$

- Higher productivity reduces the labor required for the modeled agricultural consumption component.
- At a common productivity level, higher $\bar{a}$ implies greater agricultural labor absorption.
- As A_a becomes large, ℓ_a approaches α.
- These are model allocations. They do not establish the welfare of departing workers.

▸ Worked allocation

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The calibration inputs

- 48 states and six decennial observations, 1960–2010: 288 state-years.
- IPUMS person-weighted agricultural, forestry, and fishing labor-force shares.
- Raw ERS agricultural output quantity index divided by labor input quantity index.
- Baseline recovery years: 1960 and 1970.
- One common α and a time-invariant $\bar{a}_s$ for each state.

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Step 1: recover a parameter for each candidate α

Rearrange the labor-share equation for an early observation:

$$\bar{a}_{s,t}(\alpha) = A_{a,s,t}\frac{\ell^{obs}_{a,s,t} - \alpha}{1-\alpha}.$$

Average the recovery terms across the chosen early years:

$$\bar{a}_s(\alpha) = \frac{1}{|T_{early}|}\sum_{t\in T_{early}} A_{a,s,t}\frac{\ell^{obs}_{a,s,t} - \alpha}{1-\alpha}.$$

The state parameters are conditional algebraic recoveries. They are not 48 freely optimized coefficients at this step.

▸ Worked recovery

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Step 2: select the common preference parameter

For every candidate α, form predictions for all six dates:

$$\widehat{\ell}_{a,s,t}(\alpha) = \alpha + (1-\alpha)\frac{\bar{a}_s(\alpha)}{A_{a,s,t}}.$$

Choose

$$\widehat{\alpha} = \arg\min_{\alpha \in \mathcal{A}} \sum_s \sum_t \left[\ell^{obs}_{a,s,t} - \widehat{\ell}_{a,s,t}(\alpha)\right]^2, \qquad \widehat{\bar{a}}_s = \bar{a}_s(\widehat{\alpha}).$$

Later dates help choose $\widehat{\alpha}$. They assess fit beyond the recovery moments but are not a fully held-out prediction sample.

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The profiled prediction is linear in α

Technical detail: an algebraic characterization

Define the early-period averages

$$B_s = \text{mean}_{early}(A_{a,s,t}\ell^{obs}_{a,s,t}), \qquad C_s = \text{mean}_{early}(A_{a,s,t}).$$

Then the recovered parameter and fitted share satisfy

$$\bar{a}_s(\alpha) = \frac{B_s - \alpha C_s}{1-\alpha},$$

$$\widehat{\ell}_{a,s,t}(\alpha) = \frac{B_s}{A_{a,s,t}} + \alpha\left(1 - \frac{C_s}{A_{a,s,t}}\right).$$

Substituting the recovery step removes the apparent nonlinearity in α from the prediction.

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Why a constrained optimum can lie at the bound

Technical detail

Let $d_{s,t} = 1 - C_s/A_{a,s,t}$ and $r_{s,t} = \ell^{obs}_{a,s,t} - B_s/A_{a,s,t}$. The objective becomes

$$SSE(\alpha) = \sum_{s,t} (r_{s,t} - \alpha d_{s,t})^2.$$

If $\sum d^2_{s,t} > 0$, its unconstrained minimum is

$$\alpha^{unc} = \frac{\sum_{s,t} d_{s,t} r_{s,t}}{\sum_{s,t} d^2_{s,t}}.$$

For an implemented closed interval $[a_L, a_U]$, the constrained minimizer is $\min\{a_U, \max\{a_L, \alpha^{unc}\}\}$, up to numerical tolerance.

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The positivity constraint matters

- The implemented upper bound is tied to the smallest early-year labor share, multiplied by 0.999, subject to the code's overall bound.
- This sufficient restriction keeps the early recovery terms positive.
- Rhode Island's 1970 labor share sets the baseline bound at approximately 0.00673213.
- The reported $\widehat{\alpha} = 0.00673174$ is within optimization tolerance of that bound.
- The baseline estimate is a constrained boundary solution, not an interior optimum.

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What a small $\widehat{\alpha}$ implies

$$\widehat{\bar{a}}_s = \frac{1}{|T_{early}|} \sum_{t \in T_{early}} A_{a,s,t} \frac{\ell^{obs}_{a,s,t} - \widehat{\alpha}}{1 - \widehat{\alpha}}.$$

When $\widehat{\alpha}$ is small relative to the observed labor share,

$$\widehat{\bar{a}}_s \approx \text{mean}_{t \in T_{early}} \left(A_{a,s,t} \ell^{obs}_{a,s,t} \right).$$

- This approximation is weaker for states whose labor share is close to $\widehat{\alpha}$.
- Productivity and labor absorption jointly determine the recovered object.
- Subsequent associations concern this model-generated quantity, not an independently observed entitlement measure.

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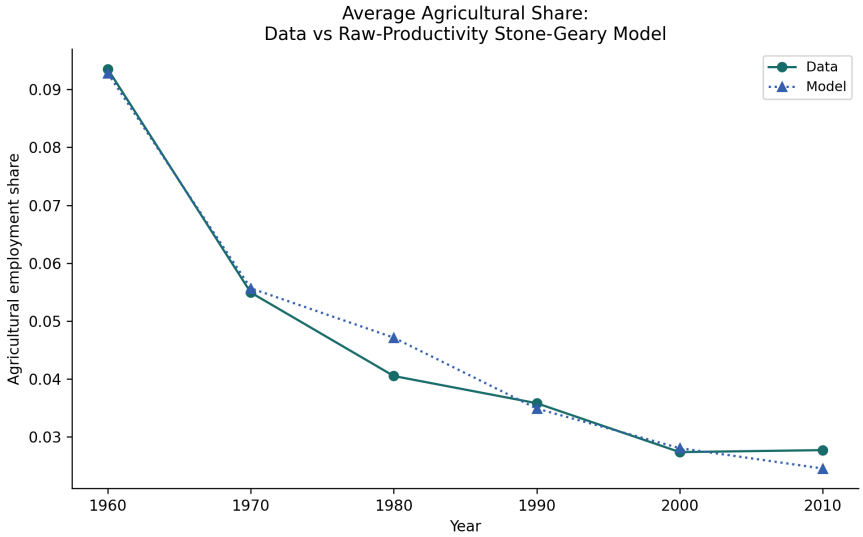
Early-window calibration results

Recovery years	1960	1960, 1970	1960, 1970, 1980
$\widehat{\alpha}$	0.0076	0.0067	0.0067
Mean $\widehat{a}_s$	0.0160	0.0162	0.0154
Median $\widehat{a}_s$	0.0114	0.0112	0.0115
Mean Soft-DTW score	90.50	90.50	90.50
Median Soft-DTW score	97.43	97.43	97.43
Mean RMSE	0.0091	0.0088	0.0086
States with score > 90	38	38	38
States / state-years	48 / 288	48 / 288	48 / 288

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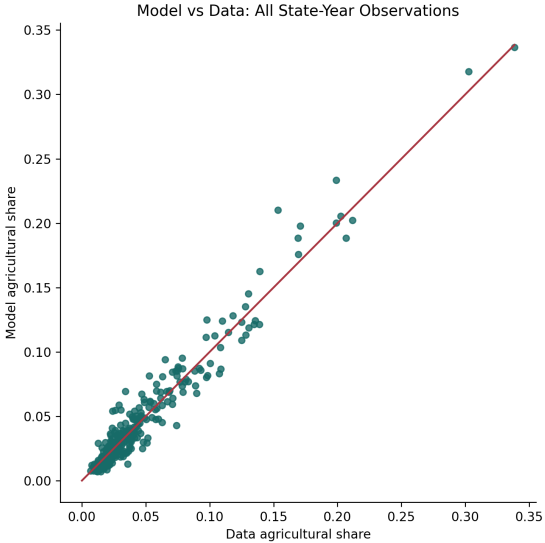
Average labor shares: data and model



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State-year labor shares: data and model



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Soft-DTW as a trajectory diagnostic

Cuturi and Blondel (2017)

- Standardize each state's observed and predicted labor-share series over time.
- Soft dynamic time warping compares the two trajectories while allowing flexible temporal alignment.
- A smoothing parameter replaces the hard minimum over alignment paths by a smooth counterpart.
- The calculation follows calibration. It does not select α or $\bar{a}_s$.

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What time warping compares

Intuition

- Ordinary point-by-point comparison matches each observed date only to the same simulated date.
- Dynamic time warping can align nearby positions when similar patterns occur at slightly different times.
- Standardization first removes each path's level and scale.
- A high trajectory score can therefore coexist with a poor match to absolute labor-share levels.

The paper reports RMSE for level fit and Soft-DTW for standardized trajectory similarity because they answer different questions.

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Soft-DTW recursion

Technical detail: the method underlying the diagnostic

For standardized series $a_1, \ldots, a_T$ and $b_1, \ldots, b_T$, let $D_{ij} = (a_i - b_j)^2$.

$$R_{ij} = D_{ij} + \operatorname{softmin}_\gamma\{R_{i-1,j}, R_{i,j-1}, R_{i-1,j-1}\},$$

$$\operatorname{softmin}_\gamma\{v_k\} = -\gamma \log\left(\sum_k e^{-v_k/\gamma}\right), \qquad \gamma > 0.$$

Set $R_{00} = 0$ and the remaining zeroth row and column to $+\infty$. The terminal value is R_{TT}. As $\gamma \downarrow 0$, the smooth minimum approaches a hard minimum. No numerical smoothing parameter is newly selected in this lecture.

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Distance normalization and its interpretation

With state distance d_s, define

$$\text{Match}_s = 100\left(1 - \frac{d_s - d_{min}}{d_{max} - d_{min}}\right).$$

- The best distance in this sample receives 100 and the worst receives zero.
- A score of 95 is not a claim of 95% accuracy or a 95% confidence level.
- A change in the reference set can affect this min–max normalization.
- Later weighting and selection use this relative measure of trajectory similarity.

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Why the trajectory scores are identical across windows

Within state s, write

$$\widehat{\ell}_{a,s,t} = \widehat{\alpha} + B_s A_{a,s,t}^{-1}, \qquad B_s = (1-\widehat{\alpha})\widehat{a}_s > 0.$$

Time standardization removes the intercept and positive scale:

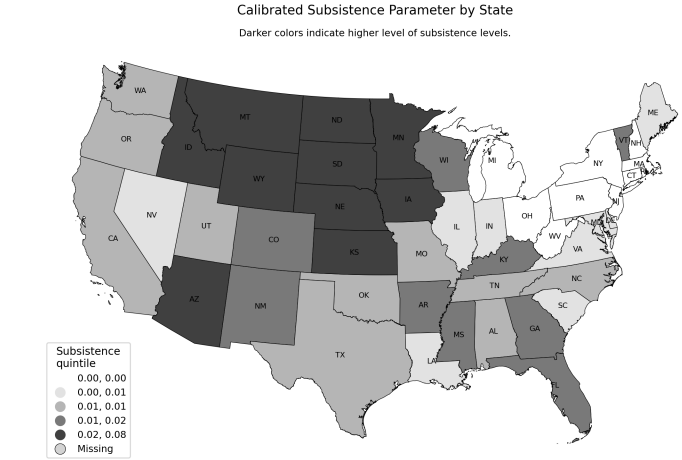
$$z_t(\widehat{\ell}_{a,s,t}) = z_t(A_{a,s,t}^{-1}).$$

The diagnostic therefore compares the shape of observed labor absorption with inverse productivity. It does not test the level or precision of the recovered $\widehat{a}_s$.

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The spatial pattern of calibrated subsistence

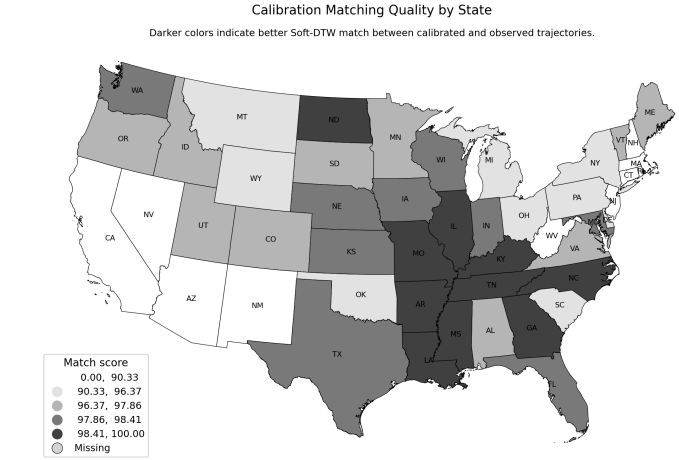


Notes: Mainland U.S. states only; Alaska and Hawaii omitted for readability. States grouped into five quantile bins. N = 48.

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The spatial pattern of trajectory similarity



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Reliability weights and selected samples

$$w_s = \mathrm{Match}_s/100, \qquad w_s^{sq} = (\mathrm{Match}_s/100)^2.$$

- Weighting emphasizes states with closer standardized trajectory alignment.
- The preferred state SNAP sample uses $\mathrm{Match}_s > 95$, leaving 33 states.
- Full-sample exercises use 48 states unless zero effective weights remove a state.
- These are fit-based robustness weights, not classical inverse-error-variance GLS weights.
- Weighting does not itself propagate uncertainty in a generated regressor.

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Historical tenure within the first slice

- The two-sector model connects agricultural labor absorption to productivity and a recovered subsistence parameter.
- The next comparison asks whether that parameter varies with historical forms of land access.
- The 1940 tenure shares summarize tenant, full-owner, and part-owner composition among farm operators.
- The cross-section concerns subsistence. The panel concerns heterogeneity in the productivity–labor relationship.

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Why tenure enters the paper

- Historical tenure describes one observable dimension of access to agricultural livelihoods.
- A tenant may depend strongly on farming while holding insecure claims to land and output.
- An owner may face debt, exclusion from credit, or dependence on commercial contracts.
- Declining tenancy can reflect agricultural exit as well as movement into ownership.
- The tenure exercises contextualize subsistence without treating tenure categories as direct measures of security.

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Historical evidence gives institutional specificity

- Depew et al. (2013) connect the AAA cotton-reduction program to displacement among sharecroppers and tenants.
- Jung (2020) relates cotton mechanization to declining tenancy and movement toward low-skill manufacturing.
- Alston and Ferrie (1999) examine the decline of paternalistic labor arrangements and changes in opposition to public welfare provision.
- These histories distinguish mechanisms that an operator-count cross-section cannot trace.

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Firsthand accounts and the meaning of agricultural access

- Nate Shaw's oral narrative recounts a confrontation over the seizure of a neighboring farmer's property and his subsequent imprisonment.
- Fannie Lou Hamer described removal from the plantation where she worked after attempting voter registration in 1962.
- Political authority could withdraw access to work and housing even after years of productive labor.
- Such evidence prevents interpreting every agricultural departure as a voluntary response to higher productivity elsewhere.

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The three tenure indicators

$$x_{sj} = \frac{\text{operators in category } j \text{ in state } s}{\text{all farm operators in state } s}, \quad j \in \{\text{tenant, full owner, part owner}\}.$$

- The data describe 1940 operator counts in 48 states.
- These are shares of operators, not shares of operated acres.
- Managers enter the denominator but not the three selected indicators.
- Consequently, the three selected shares need not sum to one.

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PCA begins with standardized observed shares

$$z_{sj} = \frac{x_{sj} - \bar{x}_j}{s_j}, \quad \bar{x}_j = \frac{1}{48}\sum_s x_{sj}, \quad s_j^2 = \frac{1}{48}\sum_s (x_{sj} - \bar{x}_j)^2.$$

- x_{sj} is an observed tenure share.
- z_{sj} is that share expressed in cross-state standard deviations.
- Standardization prevents larger original variance from automatically dominating the combination.
- The summary score combines these standardized indicators. It is a different object from any single share.

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One observed share versus a combined tenure score

A numerical example of standardization, not fitted sample moments

If a state's tenant share is 0.40, the sample mean is 0.30, and the sample standard deviation is 0.10, then

$$x_{s,tenant} = 0.40, \qquad z_{s,tenant} = \frac{0.40 - 0.30}{0.10} = 1.$$

- x answers: what fraction of operators are tenants?
- z answers: how far above or below the cross-state average is this share?
- u_s combines *three* standardized shares using estimated PCA weights.
- η_s^{PCA} standardizes that combination to one cross-state standard deviation.

The procedure summarizes composition. It does not estimate unobserved rental security or turn an operator category into a direct measure of entitlement.

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The PCA optimization problem

Let Z collect the standardized indicators and $S_Z = Z'Z/48$.

$$v_1 = \arg\max_{v'v=1} v' S_Z v, \qquad S_Z v_1 = \lambda_1 v_1.$$

- The largest-eigenvalue eigenvector gives the most variable unit-length linear combination.
- Choose its sign so that the tenant coefficient is positive.
- The score uses tenure variation only. SNAP and calibrated subsistence do not select the loadings.
- This is a descriptive dimensional reduction, not a Bayesian latent-variable estimation.

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Why PCA leads to an eigenvector

Deriving the variance-maximizing combination

The variance of Zv is $v'S_Zv$. Constrain $v'v = 1$ to prevent arbitrary rescaling:

$$\mathcal{L}(v, \lambda) = v'S_Zv - \lambda(v'v - 1).$$

The first-order condition is

$$2S_Zv - 2\lambda v = 0 \quad \Longrightarrow \quad S_Zv = \lambda v.$$

At a unit-length eigenvector, $v'S_Zv = \lambda$, so the largest eigenvalue gives the highest variance.

- Since all three indicators are standardized, $\mathrm{tr}(S_Z) = 3$.
- The reported variance fraction is $\lambda_1/3 = 69.3\%$, subject to rounding.
- Both v_1 and $-v_1$ solve the problem. The tenant-positive orientation fixes the meaning of a higher score.

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The estimated tenure combination

$$u_s \approx 0.6321 z_{s,tenant} - 0.6922 z_{s,full\ owner} + 0.3484 z_{s,part\ owner},$$

$$\eta_s^{PCA} = \frac{u_s - \bar{u}}{s_u}.$$

- The implementation uses unrounded loadings.
- The final score has cross-state mean zero and population variance one.
- Higher scores indicate a more tenant/part-owner-oriented composition relative to full ownership.
- PC1 explains 69.3% of standardized indicator variance, not 69.3% of entitlement security.

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What one tenure dimension can leave out

- Identical tenant shares can coexist with different full-owner and part-owner shares.
- Different compositions can yield the same weighted score.
- Part ownership may indicate commercial expansion or insufficient owned land.
- Contracts, credit access, and bargaining power can differ within the same category.
- The score summarizes observed composition without resolving these institutional meanings.

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Tenure and the calibrated parameter

$$\log \widehat{\bar{a}}_s = b_0 + b_1 \eta_s^{PCA} + \delta_{d(s)} + \varepsilon_s.$$

- $\delta_{d(s)}$ denotes Census-division effects, not the broader region effects used in the farm and SNAP regressions.
- The score remains fixed on its original 48-state scale.
- Specifications vary division controls, fit-based weighting, and the high-match sample.
- b_1 measures a log-index difference per one standard deviation of tenure composition.

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Tenure cross-sectional results

	All states	Division FE	FE, WLS	Match > 95
η_s^{PCA}	0.813^{***}	0.434^{***}	0.499^{***}	0.388^{**}
	(0.140)	(0.149)	(0.133)	(0.161)
Division effects	No	Yes	Yes	Yes
Fit weights	No	No	Match/100	No
States	48	48	47	33
R^2	0.466	0.681	0.716	0.735

HC1 standard errors; normal-reference tests. $^{**}p < 0.05$, $^{***}p < 0.01$. Reported R^2 is unweighted, including for WLS.

The positive association remains within divisions. One zero-weight state does not contribute to the weighted regression.

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The panel question

- The outcome is observed agricultural, forestry, and fishing labor-force share.
- Each state's 1940 tenure score is attached to its six observations from 1960 to 2010.
- The structural model motivates inverse productivity as the time-varying regressor.
- Interacting it with tenure allows historically different tenure contexts to have different labor-share slopes.

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Year effects and state effects have different roles

With year effects only,

$$\ell_{a,s,t} = b_0 + \theta A^{-1}_{a,s,t} + \phi \eta^{PCA}_s + \omega \eta^{PCA}_s A^{-1}_{a,s,t} + \lambda_t + e_{s,t}.$$

With additive state and year effects,

$$\ell_{a,s,t} = \theta A^{-1}_{a,s,t} + \omega \eta^{PCA}_s A^{-1}_{a,s,t} + \mu_s + \lambda_t + e_{s,t}.$$

The time-invariant tenure main effect is absorbed by state effects. Its interaction with changing productivity remains estimable.

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Why the tenure interaction survives state effects

Write $q_{s,t} = A_{a,s,t}^{-1}$ and first remove each state's time average:

$$\eta_s - \overline{\eta}_s = 0, \qquad \eta_s q_{s,t} - \overline{\eta_s q}_s = \eta_s (q_{s,t} - \bar{q}_s).$$

- The main effect of historical tenure is constant within each state and disappears.
- Its interaction varies whenever inverse productivity varies within the state.
- Adding year effects then removes common time components, while retaining the remaining interaction variation when the regressors have full rank.

The comparison is about different *slopes* across historical contexts, not an estimate of how a change in a state's tenant share changes its labor allocation.

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Panel estimates

	Year FE	State/year FE	State/year FE, WLS
Inverse productivity	-0.001	0.001	0.000
	(0.003)	(0.003)	(0.003)
Tenure main effect	0.007	absorbed	absorbed
	(0.005)		
Tenure $\times$ inverse productivity	0.005^{***}	0.007^{***}	0.007^{***}
	(0.001)	(0.002)	(0.002)
Observations	288	288	282
Contributing states	48	48	47
R^2	0.532	0.883	0.881

State-clustered covariance with finite-sample correction; normal-reference tests. $^{***}p < 0.01$.

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Interpreting a positive interaction

$$\frac{\partial \widehat{\ell}_a}{\partial (A_a^{-1})} = \theta + \omega \eta^{PCA}, \qquad \frac{\partial \widehat{\ell}_a}{\partial A_a} = -\frac{\theta + \omega \eta^{PCA}}{A_a^2}.$$

- $\omega > 0$ means the inverse-productivity slope increases with the tenure score.
- For a common decrease in inverse productivity, higher tenure scores imply a more negative fitted labor-share change.
- A negative productivity slope in a particular state still requires $\theta + \omega \eta_s^{PCA} > 0$.
- The model does not estimate changes in tenure itself.

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What the tenure exercises contribute

- Historically tenant-oriented states have higher calibrated subsistence in the reported cross-section.
- Their agricultural labor absorption also has a different conditional relationship to productivity.
- Dependence on agriculture can coexist with coercion or insecure access.
- The panel is an internal heterogeneity check, not independent validation of the calibration or identification of the soybean IV mechanism.

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Profit and Subsistence

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From agricultural dependence to operating control

- The first slice relates productivity, labor absorption, and historical tenure to subsistence.
- The second slice asks how operated land is distributed across farms and how that structure relates to the same calibrated index.
- Farm counts discipline a distribution of holdings. Additional equilibrium restrictions give the fitted ratio a relative-margin interpretation.
- Subsequent labor-productivity growth supplies a second association, alongside the subsistence comparison.

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The economic question

Adaptation of Klette and Kortum (2004)

- Agricultural transformation changes output per worker and who operates productive resources.
- A productivity improvement can coincide with the replacement of an existing operator.
- The model connects this process to the distribution of operated farm sizes.
- The empirical analysis asks how historical farm-size structure relates to later subsistence and productivity growth.

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The extension's place in the paper

Earlier calibration	Uses agricultural labor shares and productivity to recover the state index $\widehat{\widehat{a}}_s$.
Farm dynamics	Fits farm counts by acreage class to obtain $\widehat{p}_{s,\tau}$ and construct $\mathcal{M}_{s,\tau}$.
Empirical comparison	Relates $\mathcal{M}_{s,\tau}$ to fixed $\log \widehat{\widehat{a}}_s$ and later labor-productivity growth.
Historical interpretation	Examines productive advance alongside changes in access to agricultural livelihoods.

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The lecture sequence

1. Production, demand, limit pricing, and profits per operated unit.
2. Household discounting, incumbent optimization, and free entry.
3. Equilibrium resource allocation and productivity growth.
4. Stationary farm-size distributions and adjustment speed.
5. The conditional margin mapping and index normalization.
6. Historical calibration, empirical associations, and interpretation.

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Notation: production and value

$s,\ t,\ \tau$	State, continuous model time, and census snapshot date.
$j \in [0,1]$	A differentiated productive unit. Aggregate unit mass equals one.
$n \geq 1$	Number of productive units operated by a farm.
$q_{sj}, y_{sj}, \ell_{sj}$	Unit productivity, output, and production labor.
Y_s, w_s, r_s	Final output, wage, and market interest rate.
Λ_s, π_s	Quality step/price–cost ratio and gross operating margin.
$V_s(n,t), \nu_s, v_s$	Firm value, current unit value, and unit value divided by Y_s.

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Notation: dynamics and empirical objects

u, z_s	Chosen per-unit expansion intensity and its equilibrium value.
x_s, e_s	Per-unit displacement hazard and entrant-farm flow.
m, b	Units per entrant and acres represented by one unit.
$\beta_I, c_{I,s}, c_{E,s}$	Innovation elasticity and labor-cost parameters.
$p_s = z_s/x_s$	Expansion relative to displacement.
μ_n, F, f_n	Farm mass at size n, total farm mass, and farm-count probability.
$\mathcal{M}_{s,\tau}$	Standardized fitted inverse rate ratio.
$\widehat{\bar{a}}_s, G_s^A$	Calibrated subsistence index and observed productivity growth.

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Productive units and operated acreage

- The economy has a fixed productive resource with aggregate measure one.
- Farms differ in the number of units they operate.
- Operating control can differ from legal ownership.
- The empirical application maps one unit to b acres, with baseline $b = 1$.
- This acreage mapping does not establish that acres are identical technologies or distinct product markets.

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Closed-resource reallocation

Assumption 4.1: expansion changes control, not the resource total

$$A_s(t) = \sum_{n \geq 1} n\mu_{s,n}(t) = 1, \qquad \dot{A}_s(t) = 0.$$

Summing operated-unit gains and losses gives

$$\dot{A}_s = \underbrace{z_s A_s}_{\text{incumbent acquisitions}} + \underbrace{me_s}_{\text{entrant acquisitions}} - \underbrace{x_s A_s}_{\text{incumbent losses}}.$$

Thus $x_s = z_s + me_s$: every acquired position displaces an existing operator, without compensation in the maintained model.

- Productivity can grow on the fixed resource through replacement.
- Conservation holds even while the distribution of holdings changes; stationarity is an additional restriction.

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Expansion, replacement, and exit

For a farm currently operating n units, over a short interval dt:

$$\begin{aligned}
&\text{Acquire one unit:} && n \mapsto n+1, && nu\,dt + o(dt),\\
&\text{Lose one unit:} && n \mapsto n-1, && nx_s\,dt + o(dt),\\
&\text{No size change:} && n \mapsto n, && 1 - n(u+x_s)dt + o(dt).
\end{aligned}$$

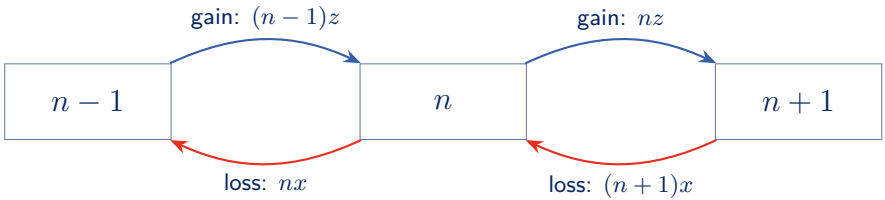
- Losing the last unit ends the incumbent's operation.
- A new entrant begins with m existing units.
- Successful replacement raises unit productivity by $\Lambda_s > 1$.

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A farm moves between neighboring sizes

The stochastic process behind the stationary distribution



- The diagram shows equilibrium incumbent rates. A choosing farm replaces z by its own control u for acquisition.
- A farm at size one exits if it loses its last unit; size zero is absorbing.
- New farms enter at m units. Existing productive units change operators; the model does not create additional land.

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A transition rate is not an annual probability

Illustrative numbers, not calibrated annual rates

Suppose a farm currently operates $n = 100$ units and chooses $u = 0.02$ per year. Its instantaneous acquisition intensity is $nu = 2$ events per year.

$$\Pr(\text{one acquisition in } dt \mid n = 100) = 2\,dt + o(dt).$$

- Over a very short interval, the rate times the interval approximates an event probability.
- It is not correct to call $nu = 2$ a probability: probabilities cannot exceed one.
- Each gain or loss changes n, so subsequent event intensities change too.

For a *fixed* Poisson intensity of two, the probability of at least one event over a year would be $1 - e^{-2}$, not two. The farm process has size-dependent intensities.

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Production and competition assumptions

1. A competitive final-good sector combines differentiated inputs through a symmetric Cobb–Douglas aggregator.
2. Intermediate production uses labor linearly.
3. The preceding technology remains a competitive threat.
4. All units in one environment have the same quality step Λ_s.
5. The final-good price is the numeraire.

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Reallocation assumptions

- Innovation uses labor and existing units through a constant-returns technology.
- Expansion and loss hazards are proportional to current farm size.
- Targets are uniform across the fixed productive resource.
- Entry costs and entry values are linear in the initial holding m.
- Transfers involve no acquisition payment, compensation, or salvage value in this model.

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Households and local equilibrium closure

- Households supply a fixed labor resource L_s.
- Diversified owners hold firms and value cash flows at the market rate r_s.
- Preferences are logarithmic in consumption.
- Aggregate uncertainty, interstate trade, and interstate labor mobility are excluded.
- Innovation and entry use labor directly. Final output can therefore all be consumed.

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The additional restriction behind relative margins

$$\frac{c_{E,s}}{L_s} = f_E > 0, \qquad \frac{c_{I,s}}{L_s} = f_I > 0.$$

- ρ, β_I, f_E, and f_I are common across states and both census environments.
- Costs are common relative to the state's model labor resource.
- Acreage counts do not establish this restriction.
- It supplies a common mapping from p^{-1} to operating margins.

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The final-good producer's problem

Normalize the price of final output to one. The producer solves

$$\sup_{\{y_{sj}>0\}} \left\{ \mathcal{F}_s(\boldsymbol{y}_s) - \int_0^1 P_{sj} y_{sj}\, dj \right\},$$

where

$$Y_s = \mathcal{F}_s(\boldsymbol{y}_s) = \exp\left\{ \int_0^1 \log y_{sj}\, dj \right\}.$$

- Inactivity is feasible.
- Constant returns determine the input mix at given scale.
- Resource clearing will determine aggregate output.

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Input demand: differentiation step by step

Because $\log \mathcal{F}_s = \int_0^1 \log y_{sj}\, dj$,

$$\frac{\partial \mathcal{F}_s}{\partial y_{sj}} = \mathcal{F}_s \frac{1}{y_{sj}} = \frac{Y_s}{y_{sj}}.$$

The first-order condition equates marginal product to input price:

$$\frac{Y_s}{y_{sj}} - P_{sj} = 0 \quad \Longrightarrow \quad y_{sj} = \frac{Y_s}{P_{sj}}.$$

Consequently,

$$P_{sj} y_{sj} = Y_s, \qquad -\frac{\partial \log y_{sj}}{\partial \log P_{sj}} = 1.$$

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The zero-profit price index

Substitute $y_{sj} = Y_s/P_{sj}$ into the aggregator:

$$\begin{aligned}\log Y_s &= \int_0^1 \left(\log Y_s - \log P_{sj}\right) dj \\ &= \log Y_s - \int_0^1 \log P_{sj}\, dj.\end{aligned}$$

Hence positive production requires

$$\int_0^1 \log P_{sj}\, dj = 0, \qquad c_s^F(\boldsymbol{P}_s) = \exp\left\{\int_0^1 \log P_{sj}\, dj\right\} = 1.$$

The competitive final-good price equals unit cost.

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Intermediate production and marginal cost

An operator with productivity q_{sj} solves

$$\min_{\ell \geq 0} w_s \ell \quad \text{subject to} \quad q_{sj}\ell \geq \bar{y}.$$

Positive wages make the production requirement bind:

$$\ell^* = \frac{\bar{y}}{q_{sj}}, \qquad \mathcal{C}^I_{sj}(\bar{y}) = \frac{w_s}{q_{sj}}\bar{y}, \qquad MC_{sj} = \frac{w_s}{q_{sj}}.$$

Higher q_{sj} lowers the labor required for each unit of output.

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The competitive threat

The current operator's predecessor has productivity q_{sj}/Λ_s.

$$MC_{sj}^{\text{old}} = \frac{w_s}{q_{sj}/\Lambda_s} = \Lambda_s MC_{sj}.$$

- The predecessor can supply at this cost.
- The current operator retains the market only at a price no higher than this ceiling.
- Ties favor the more productive operator.

Thus its relevant price interval is

$$MC_{sj} \leq P \leq \Lambda_s MC_{sj}.$$

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The intermediate producer's optimal price

Using final-sector demand, the producer solves

$$\max_{MC_{sj} \le P \le \Lambda_s MC_{sj}} (P - MC_{sj}) \frac{Y_s}{P} = \max_P Y_s \left(1 - \frac{MC_{sj}}{P}\right).$$

The derivative is strictly positive:

$$\frac{d}{dP}\left[Y_s\left(1 - \frac{MC_{sj}}{P}\right)\right] = \frac{Y_s MC_{sj}}{P^2} > 0.$$

The ceiling therefore binds:

$$P_{sj} = \Lambda_s MC_{sj} = \frac{\Lambda_s w_s}{q_{sj}}.$$

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Markup, margin, and profit flow

Three related quantities have different units:

$$\text{Price–cost ratio:} \quad \Lambda_s = \frac{P_{sj}}{MC_{sj}},$$

$$\text{Gross operating margin:} \quad \pi_s = \frac{P_{sj} - MC_{sj}}{P_{sj}} = 1 - \Lambda_s^{-1},$$

$$\text{Profit per operated unit:} \quad \Pi_{sj}^{I} = \pi_s Y_s.$$

- A margin is a fraction of revenue.
- Profit flow depends on both the margin and aggregate expenditure.
- These profits exclude production labor costs, but omit rent, debt service, and other unmodeled costs.

▸ Markup example

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Why unit quality disappears from operating profits

Substitute the optimal price into demand and technology:

$$y_{sj} = \frac{Y_s q_{sj}}{\Lambda_s w_s}, \qquad \ell_{sj} = \frac{y_{sj}}{q_{sj}} = \frac{Y_s}{\Lambda_s w_s},$$

$$P_{sj} y_{sj} = Y_s, \qquad w_s \ell_{sj} = \frac{Y_s}{\Lambda_s}.$$

Therefore

$$\Pi^I_{sj} = Y_s - \frac{Y_s}{\Lambda_s} = \pi_s Y_s.$$

Quality affects physical output and price. Equal expenditure and a common markup make unit labor costs and profits independent of quality.

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The aggregate quality index and the wage

Define

$$Q_s(t) = \exp\left\{\int_0^1 \log q_{sj}(t)\, dj\right\}.$$

The price-index condition gives

$$\begin{aligned} 0 &= \int_0^1 \log\left(\frac{\Lambda_s w_s}{q_{sj}}\right) dj \\ &= \log \Lambda_s + \log w_s - \log Q_s. \end{aligned}$$

Thus

$$w_s = \frac{Q_s}{\Lambda_s}.$$

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Aggregate production uses production labor

Integrate unit labor demands over the resource:

$$L_s^P = \int_0^1 \ell_{sj}\, dj = \frac{Y_s}{\Lambda_s w_s} = \frac{Y_s}{Q_s}.$$

Consequently,

$$Y_s = Q_s L_s^P.$$

- L_s^P includes only labor producing intermediate outputs.
- Innovation and entry also use labor.
- The model does not equate Q_s to the observed ERS output/labor index.

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The household problem

Given prices and initial financial wealth, the household solves

$$\max_{\{C_s(t)>0\}} \int_0^\infty e^{-\rho t} \log C_s(t)\, dt$$

subject to

$$\dot{B}_s = r_s B_s + w_s L_s - C_s, \qquad \rho > 0,$$

and a no-Ponzi condition.

- B_s is diversified financial wealth, including firm equity.
- Labor supply is inelastic.
- r_s is the market return, while ρ is time preference.

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The household's current-value Hamiltonian

$$\mathcal{H}_s^H = \log C_s + \psi_s(r_s B_s + w_s L_s - C_s).$$

The consumption and costate conditions are

$$\frac{1}{C_s} - \psi_s = 0, \qquad \dot{\psi}_s = \rho\psi_s - \frac{\partial \mathcal{H}_s^H}{\partial B_s} = (\rho - r_s)\psi_s.$$

Differentiate $\psi_s = C_s^{-1}$:

$$\frac{\dot{\psi}_s}{\psi_s} = -\frac{\dot{C}_s}{C_s} \quad \Longrightarrow \quad \frac{\dot{C}_s}{C_s} = r_s - \rho.$$

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Growth-adjusted discounting

All final output is consumed, so $C_s = Y_s$. On a balanced-growth path,

$$\frac{\dot{C}_s}{C_s} = \frac{\dot{Y}_s}{Y_s} = g_s, \qquad r_s - g_s = \rho.$$

Constant Λ_s and constant labor allocations also imply

$$\frac{\dot{w}_s}{w_s} = g_s, \qquad \omega_s \equiv \frac{w_s}{Y_s} \text{ is constant.}$$

Dividing a growing firm's value by output will replace r_s with the growth-adjusted rate ρ.

▸ CRRA comparison

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General innovation technology

A farm uses innovation labor R and its existing n units:

$$Z = \left(\frac{R}{c_{I,s}}\right)^{\beta_I} n^{1-\beta_I}, \qquad 0 < \beta_I < 1.$$

Define $u = Z/n$. Then

$$un = \left(\frac{R}{c_{I,s}}\right)^{\beta_I} n^{1-\beta_I},$$
$$u = \left(\frac{R}{nc_{I,s}}\right)^{\beta_I},$$
$$R = nc_{I,s}u^{1/\beta_I}.$$

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Convex innovation costs

Innovation expenditure is

$$w_s R = w_s n c_{I,s} u^{1/\beta_I}.$$

Per-unit marginal expenditure is

$$\frac{d}{du}\left(w_s c_{I,s} u^{1/\beta_I}\right) = \frac{w_s c_{I,s}}{\beta_I} u^{(1-\beta_I)/\beta_I}.$$

- More intensive expansion becomes increasingly costly.
- At a fixed intensity, total expenditure scales linearly with n.
- $\beta_I = 1/2$ gives quadratic costs, but the derivation keeps β_I general.

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Aggregate replacement accounting

With $\sum_n n\mu_{sn} = 1$, incumbent acquisitions occur at total flow

$$\sum_{n\geq 1} nz_s\mu_{sn} = z_s.$$

Entrants acquire existing units at total flow me_s. Uniform targeting gives

$$x_s = z_s + me_s.$$

Therefore

$$p_s = \frac{z_s}{x_s}, \qquad 1 - p_s = \frac{me_s}{x_s}, \qquad p_s^{-1} = 1 + \frac{me_s}{z_s}.$$

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A rate ratio and three different entry measures

$p = z/x$	Incumbent acquisitions relative to total unit displacement.
$1 - p = me/x$	Entrant-acquired units as a share of displaced units.
e	Entrant firms per unit of the aggregate resource per unit of time.
e/F	Entrant firms relative to the incumbent farm mass per unit of time.

Neither p nor $1 - p$ alone identifies an annual farm-entry frequency.

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The incumbent's dynamic objective

The incumbent chooses an admissible nonnegative intensity process:

$$V_s(n,t) = \sup_{\{u(a)\geq 0\}} \mathbb{E}_{n,t}\left[\int_t^\infty e^{-\int_t^a r_s(v)\,dv} n(a) \times \left\{\pi_s Y_s(a) - w_s(a) c_{I,s} u(a)^{1/\beta_I}\right\} da\right].$$

- Operating profits and innovation costs accrue while the farm survives.
- Farm size follows a stochastic jump process.
- Exit is absorbing, with $V_s(0,t) = 0$.

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The short-interval Bellman recursion

Suppress s and define $\varphi(n,u,t) = n[\pi Y - wc_I u^{1/\beta_I}]$.

$$V(n,t) = \max_{u \geq 0} \Big\{ \varphi(n,u,t)\,dt + (1 - r\,dt)\big[(1 - n(u+x)dt)V(n,t+dt) + nu\,dt\,V(n+1,t+dt) + nx\,dt\,V(n-1,t+dt)\big] \Big\} + o(dt).$$

Continuation value averages over no jump, one acquisition, and one loss.

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Deriving the HJB

Expand $V(n, t+dt) = V(n,t) + V_t(n,t)dt + o(dt)$. Subtract $V(n,t)$, divide by dt, and take the limit:

$$r_s V_s(n,t) - V_{s,t}(n,t) = \max_{u \geq 0} \Big\{ n\pi_s Y_s - w_s c_{I,s} n u^{1/\beta_I} + nu[V_s(n+1,t) - V_s(n,t)] + nx_s[V_s(n-1,t) - V_s(n,t)] \Big\}.$$

Each jump term equals its event intensity times its change in value.

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Reading the HJB as an economic balance

The Bellman equation compares holding a farm asset with the market return on its value:

$$\underbrace{rV}_{\text{required return}} = \underbrace{V_t}_{\text{growth at a fixed size}} + \underbrace{\text{current net profit}}_{\text{cash flow}} + \underbrace{\text{expected jump gains}}_{\text{changes in operating control}}.$$

- A gain occurs at rate nu and changes value by $V(n+1) - V(n)$.
- A loss occurs at rate nx and changes value by $V(n-1) - V(n)$.
- The farm chooses u to balance the extra expected gain against innovation expenditure.

The jump terms are expected changes in asset value, not additional sales revenue.

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Why a linear value function is plausible

Consider

$$V_s(n,t) = n\nu_s(t) = nv_sY_s(t),$$

with constant normalized unit value v_s on balanced growth.

- Every operated unit generates the same profit flow.
- Innovation expenditure at a given intensity is proportional to n.
- Acquisition and displacement rates are proportional to n.
- There are no fixed operating costs or size-dependent compensation terms.

The HJB and transversality condition will verify this candidate.

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Value increments and normalization

The candidate implies

$$V_s(n+1,t) - V_s(n,t) = v_s Y_s, \qquad V_s(n-1,t) - V_s(n,t) = -v_s Y_s.$$

Also $V_{s,t} = n v_s g_s Y_s$. Divide the HJB by nY_s:

$$(r_s - g_s)v_s = \max_{u \geq 0} \left\{ \pi_s - \omega_s c_{I,s} u^{1/\beta_I} + (u - x_s)v_s \right\}.$$

Since $r_s - g_s = \rho$,

$$\rho v_s = \max_{u \geq 0} \left\{ \pi_s - \omega_s c_{I,s} u^{1/\beta_I} + (u - x_s)v_s \right\}.$$

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Optimal incumbent expansion

Differentiate the normalized objective with respect to u:

$$-\frac{\omega_s c_{I,s}}{\beta_I} u^{(1-\beta_I)/\beta_I} + v_s = 0.$$

At the symmetric optimum $u = z_s$,

$$z_s = \left(\frac{\beta_I v_s}{\omega_s c_{I,s}}\right)^{\beta_I/(1-\beta_I)}.$$

For $u > 0$, the second derivative is

$$-\frac{\omega_s c_{I,s}(1-\beta_I)}{\beta_I^2} u^{1/\beta_I - 2} < 0.$$

Positive unit value therefore gives a unique positive optimum.

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Why displacement is held fixed in the firm's choice

Individual optimization and aggregate consistency are separate steps

The individual farm solves

$$\max_{u \geq 0} \left\{ \pi - \omega c_I u^{1/\beta_I} + (u - x)v \right\},$$

taking x, ω, and the value of a unit as given.

1. First derive the best response u^* to the aggregate environment.
2. In a symmetric equilibrium, impose $u^* = z$ for all incumbents.
3. Then impose $x = z + me$ so that aggregate acquisitions equal losses.

Substituting $x = u + me$ *before* differentiating would incorrectly make one infinitesimal farm internalize the whole economy's displacement hazard.

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Simplifying the optimal value equation

Multiply the first-order condition by $\beta_I z_s$:

$$\omega_s c_{I,s} z_s^{1/\beta_I} = \beta_I z_s v_s.$$

Substitute into the HJB:

$$\rho v_s = \pi_s - \beta_I z_s v_s + (z_s - x_s) v_s,$$
$$\pi_s = v_s \big[\rho + x_s - (1 - \beta_I) z_s\big].$$

Equivalently,

$$v_s = \frac{\pi_s - \omega_s c_{I,s} z_s^{1/\beta_I}}{\rho + x_s - z_s}.$$

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Free entry and its complementarity condition

An entrant acquires m units. Its net value is

$$V_s(m,t) - mw_s c_{E,s} = mY_s(v_s - \omega_s c_{E,s}).$$

Free entry requires

$$e_s \geq 0, \qquad v_s - \omega_s c_{E,s} \leq 0, \qquad e_s(v_s - \omega_s c_{E,s}) = 0.$$

In the maintained positive-entry regime,

$$v_s = \omega_s c_{E,s}, \qquad \nu_s = w_s c_{E,s}.$$

The factor m cancels because value and entry cost both scale linearly with initial size.

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What free entry does and does not eliminate

$$\underbrace{mv}_{\text{value of the new farm}} = \underbrace{mwc_E}_{\text{labor cost of establishing it}}.$$

- Free entry eliminates a positive *net entry opportunity*, not the gross profit earned on every operating unit.
- Those operating profits support the asset value that compensates the entrant for its establishment cost.
- The entry holding m cancels here because value and establishment cost are both linear in initial size.
- It does not disappear from the distribution: entrants are still injected at $n = m$.

The coexistence of positive operating margins and zero net entry value is therefore consistent.

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Expansion after imposing free entry

Insert $v_s = \omega_s c_{E,s}$ into the incumbent policy:

$$z_s = \left(\frac{\beta_I c_{E,s}}{c_{I,s}} \right)^{\beta_I/(1-\beta_I)}.$$

The margin equation becomes

$$\pi_s = \omega_s c_{E,s} \left[\rho + x_s - (1-\beta_I) z_s \right].$$

- The cost ratio determines equilibrium incumbent intensity.
- Entry size m affects entrant flow and the stationary size distribution.
- It does not enter this policy equation under linear entry value and cost.

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Verification and the terminal-value condition

Under $u = z_s$, expected size obeys

$$\mathbb{E}_t[n(T)] = n(t)e^{(z_s - x_s)(T-t)}.$$

Balanced growth and $r_s - g_s = \rho$ give

$$e^{-r_s(T-t)}\mathbb{E}_t[V_s(n(T), T)] = n(t)v_s Y_s(t)e^{-[\rho + x_s - z_s](T-t)} \longrightarrow 0.$$

The HJB inequality bounds admissible payoffs by the candidate value. The maximizing policy attains that value as the terminal term vanishes.

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The incumbent's current-value Hamiltonian

For a deterministic, size-independent intensity, define $N_s(t) = \mathbb{E}[n(t)]$.

$$\dot{N}_s = (u - x_s)N_s.$$

The reduced Hamiltonian is

$$\mathcal{H}_s(N_s, u, \nu_s, t) = N_s \left[\pi_s Y_s - w_s c_{I,s} u^{1/\beta_I} + \nu_s(u - x_s)\right].$$

Its control condition is

$$\nu_s = \frac{w_s c_{I,s}}{\beta_I} u^{(1-\beta_I)/\beta_I}.$$

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Why the Hamiltonian is a reduced representation

The stochastic state is the integer $n(t)$. Its expectation is $N(t) = \mathbb{E}[n(t)]$. Under a common per-unit policy,

$$\dot{N} = (u - x)N, \qquad \mathcal{H} = N\left[\pi Y - wc_I u^{1/\beta_I} + \nu(u - x)\right].$$

- Linear profits and a linear value function allow expected size to summarize expected payoffs.
- The costate ν is the current value of an additional unit, not a present-value quantity with discounting already attached.
- The costate equation uses the market rate r. The rate $\rho = r - g$ appears only after balanced-growth normalization.

Without the linear reduction, replacing random farm size by its expectation would generally not preserve the optimization problem.

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The costate equation and the HJB agree

The current-value costate equation uses the market rate r_s:

$$\dot{\nu}_s = r_s \nu_s - \left[\pi_s Y_s - w_s c_{I,s} u^{1/\beta_I} + \nu_s (u - x_s)\right].$$

Set $u = z_s$ and $\dot{\nu}_s = g_s \nu_s$, then divide by Y_s:

$$(r_s - g_s) v_s = \pi_s - \omega_s c_{I,s} z_s^{1/\beta_I} + (z_s - x_s) v_s.$$

Using $r_s - g_s = \rho$ recovers the reduced HJB.

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Equilibrium definition: optimality

Given the primitives and an initial quality distribution, a positive-entry balanced-growth equilibrium requires:

1. Household utility maximization and the budget/transversality conditions.
2. Final-good profit maximization and intermediate cost minimization and limit pricing.
3. Incumbent optimization with $V_s(0,t) = 0$ and common positive intensity z_s.
4. Positive entry satisfying $V_s(m,t) = mw_s c_{E,s}$.

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Equilibrium definition: clearing and growth

Market clearing requires

$$
\begin{aligned}
C_s &= Y_s, \\
L_s &= L_s^P + c_{I,s} z_s^{1/\beta_I} + m c_{E,s} e_s, \\
\sum_{n \geq 1} n \mu_{sn} &= 1, \qquad B_s = \sum_{n \geq 1} \mu_{sn} V_s(n,t), \\
x_s &= z_s + m e_s.
\end{aligned}
$$

Farm masses are stationary. Output, wages, and unit values grow at g_s, while labor allocations and transition intensities remain constant.

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Three uses of labor

$$\text{Production:} \quad L_s^P = \frac{Y_s}{\Lambda_s w_s} = \frac{1-\pi_s}{\omega_s},$$

$$\text{Incumbent innovation:} \quad L_s^I = c_{I,s} z_s^{1/\beta_I} = \beta_I c_{E,s} z_s,$$

$$\text{Entry:} \quad L_s^E = m c_{E,s} e_s = c_{E,s}(x_s - z_s).$$

The innovation equality uses its first-order condition and free entry. The entry equality uses replacement accounting.

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Labor clearing: the cancellation

Retain entry labor as $mc_{E,s}e_s$ until imposing conservation:

$$\begin{aligned} L_s &= \frac{1}{\omega_s} - c_{E,s}\left[\rho + x_s - (1-\beta_I)z_s\right] \\ &\quad + \beta_I c_{E,s} z_s + mc_{E,s}e_s \\ &= \frac{1}{\omega_s} - \rho c_{E,s} + c_{E,s} \underbrace{\left(z_s + me_s - x_s\right)}_{\text{zero by resource conservation}} . \end{aligned}$$

Hence

$$\omega_s = \frac{1}{L_s + \rho c_{E,s}}.$$

Fixed-resource accounting removes the reallocation-rate term. Free entry, convex labor costs, and log utility provide the other equilibrium restrictions.

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The inverse rate ratio in the margin equation

Use $x_s = z_s/p_s$ and the normalized wage:

$$\begin{aligned}\pi_s &= \omega_s c_{E,s}\left[\rho + \frac{z_s}{p_s} - (1-\beta_I)z_s\right] \\ &= \frac{c_{E,s}\{\rho + z_s[p_s^{-1} - (1-\beta_I)]\}}{L_s + \rho c_{E,s}}.\end{aligned}$$

- The distribution will identify p_s under the fitted family.
- Costs and preferences still enter the margin equation.
- Common normalized costs will later allow a relative comparison.

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Solving a symbolic equilibrium

Given $(L_s, c_{I,s}, c_{E,s}, \rho, \Lambda_s, \beta_I, m)$:

$$\pi_s = 1 - \Lambda_s^{-1}, \qquad z_s = \left(\frac{\beta_I c_{E,s}}{c_{I,s}}\right)^{\beta_I/(1-\beta_I)},$$

$$\omega_s = \frac{1}{L_s + \rho c_{E,s}}, \qquad v_s = \omega_s c_{E,s},$$

$$x_s = \frac{\pi_s}{\omega_s c_{E,s}} - \rho + (1-\beta_I) z_s, \qquad e_s = \frac{x_s - z_s}{m},$$

$$p_s = \frac{z_s}{x_s}.$$

These formulas characterize an equilibrium when the positive-entry feasibility conditions hold.

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The positive-entry condition

Since $e_s = (x_s - z_s)/m$, positive entry requires $x_s > z_s$.

$$\frac{\pi_s}{\omega_s c_{E,s}} - \rho + (1 - \beta_I) z_s > z_s,$$

$$\pi_s > \omega_s c_{E,s} (\rho + \beta_I z_s).$$

- This ensures $0 < p_s < 1$ and $e_s > 0$.
- Primitives that fail this inequality lie outside the regime analyzed here.
- $\Lambda_s > 1$ ensures a positive margin below one.

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Quality growth from replacement events

Let $N_{sj}(t)$ count replacements on a productive unit.

$$q_{sj}(t) = q_{sj}(0)\Lambda_s^{N_{sj}(t)}.$$

Each event adds $\log \Lambda_s$ to log quality. Its intensity is x_s:

$$\mathbb{E}[d \log q_{sj}] = x_s \log \Lambda_s \, dt.$$

Without aggregate uncertainty, aggregation gives

$$g_s = \frac{d \log Q_s}{dt} = x_s \log \Lambda_s, \qquad Q_s(t) = Q_s(0)e^{g_s t}.$$

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Balanced-growth paths

Once hazards and margins are determined,

$$w_s(t) = \frac{Q_s(t)}{\Lambda_s}, \qquad Y_s(t) = \frac{w_s(t)}{\omega_s} = Q_s(t) L_s^P,$$
$$C_s(t) = Y_s(t), \qquad \nu_s(t) = w_s(t) c_{E,s},$$
$$r_s = \rho + g_s, \qquad L_s^P = \frac{L_s + \rho c_{E,s}}{\Lambda_s}.$$

- Output and unit values grow even though the farm-size distribution is stationary.
- Stationarity concerns the distribution of operating control.
- The initial quality distribution sets the level $Q_s(0)$.

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Goods and financial accounting

Wages plus operating profits, less innovation and entry expenditure, satisfy

$$\begin{aligned} w_s L_s + \pi_s Y_s - w_s L_s^I - w_s L_s^E &= w_s L_s^P + \pi_s Y_s \\ &= (1 - \pi_s) Y_s + \pi_s Y_s = Y_s. \end{aligned}$$

Total equity is

$$B_s = \sum_n \mu_{sn} n \nu_s = \nu_s.$$

Thus the household budget implies

$$\dot{B}_s = r_s \nu_s + w_s L_s - Y_s = g_s \nu_s.$$

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Farm masses and farm probabilities

Let μ_n be the mass of farms with n operated units, per unit of the aggregate resource.

$$\sum_{n\geq 1} n\mu_n = 1, \qquad F = \sum_{n\geq 1} \mu_n.$$

Then

$$f_n = \frac{\mu_n}{F} \quad \text{is a farm-count probability,} \qquad a_n = n\mu_n \quad \text{is an operated-unit share.}$$

- Sampling a farm uses f_n.
- Sampling an operated unit uses a_n and puts more weight on large farms.

▸ Count and land shares

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Stationarity does not mean that farms stop changing

A stationary histogram has constant mass in each size category even though individual farms gain units, lose units, enter, and exit.

- Some farms leave a size class by expanding; others enter it by contracting.
- Entry replenishes the population at size m.
- Exit occurs when a one-unit farm loses its last unit.

stationary mass in a class $\iff$ its total inflow equals its total outflow.

Fitting this distribution is not evidence that historical agriculture actually completed its adjustment to a steady state.

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Stationary inflows and outflows

For $n \geq 1$, stationarity requires

$$0 = \underbrace{z(n-1)\mu_{n-1}}_{\text{expansion from } n-1} + \underbrace{x(n+1)\mu_{n+1}}_{\text{loss from } n+1} - \underbrace{(z+x)n\mu_n}_{\text{departures from } n} + \underbrace{e\mathbf{1}\{n=m\}}_{\text{entry at } m}.$$

The first term is zero at $n = 1$. Loss from $n = 1$ causes exit.

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A useful flow transformation

Set

$$a_n = n\mu_n, \qquad a_0 = 0, \qquad J_n = xa_n - za_{n-1}.$$

Then

$$\begin{aligned} J_{n+1} - J_n &= xa_{n+1} - za_n - xa_n + za_{n-1} \\ &= xa_{n+1} - (z+x)a_n + za_{n-1}. \end{aligned}$$

The stationary equation reduces to

$$J_{n+1} - J_n + e\mathbf{1}\{n = m\} = 0.$$

J_n measures net downward flow across the boundary between $n-1$ and n.

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Why the stationary tail has zero net flow

For $n > m$, $J_n = J$ is constant, so

$$a_n = p a_{n-1} + \frac{J}{x}, \qquad p = \frac{z}{x} < 1.$$

The recursion converges to

$$a_n \longrightarrow \frac{J}{x - z}.$$

- If $J > 0$, then $\sum_n a_n$ diverges.
- If $J < 0$, masses eventually become negative.
- Nonnegativity and finite unit mass therefore require $J = 0$.

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The lower part of the distribution

At the entry size, $J_{m+1} - J_m + e = 0$ and $J_{m+1} = 0$. Thus $J_n = e$ for $1 \leq n \leq m$.

$$\begin{aligned} a_n &= p a_{n-1} + \frac{e}{x} \\ &= \frac{e}{x} \sum_{j=0}^{n-1} p^j = \frac{e}{x} \frac{1-p^n}{1-p}. \end{aligned}$$

Since $x = z + me$,

$$\frac{e}{x} = \frac{1-p}{m}, \qquad a_n = \frac{1-p^n}{m}, \quad 1 \leq n \leq m.$$

▸ Recursion check

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The tail and the complete stationary solution

For $n > m$, zero net flow gives $a_n = pa_{n-1}$. Therefore

$$a_n = \frac{1 - p^m}{m} p^{n-m}, \qquad n > m.$$

Divide by n to obtain farm masses:

$$\mu_n(p, m) = \begin{cases} \dfrac{1 - p^n}{mn}, & 1 \le n \le m, \\ \dfrac{(1 - p^m)p^{n-m}}{mn}, & n > m. \end{cases}$$

The recursion and the tail condition uniquely determine this solution.

▸ Numerical distribution

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Checking the resource normalization

Directly summing operated-unit shares gives

$$\sum_{n\geq 1} n\mu_n = \frac{1}{m}\left[\sum_{n=1}^{m}(1-p^n) + (1-p^m)\sum_{j=1}^{\infty} p^j\right]$$
$$= \frac{1}{m}\left[m - \frac{p(1-p^m)}{1-p} + \frac{p(1-p^m)}{1-p}\right] = 1.$$

The fixed aggregate resource emerges exactly from entry and replacement accounting.

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The single-unit-entry special case

At $m = 1$,

$$\mu_n = (1-p)\frac{p^{n-1}}{n}.$$

Using $\sum_{n\geq 1} p^n/n = -\log(1-p)$,

$$F(p,1) = -\frac{1-p}{p}\log(1-p), \qquad f_n = -\frac{p^n}{n\log(1-p)}.$$

This is the logarithmic farm-count distribution. Multi-unit entry changes the lower distribution while retaining proportional expansion, displacement, and exit.

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Farm shares, land shares, and mean acreage

For a class of sizes $\mathcal{B}_k$,

$$\text{Farm share} = \frac{\sum_{n\in\mathcal{B}_k} \mu_n}{F}, \qquad \text{Land share} = \sum_{n\in\mathcal{B}_k} n\mu_n.$$

Mean farm size is

$$\mathbb{E}_f[n] = \sum_n n\frac{\mu_n}{F} = \frac{1}{F}, \qquad \mathbb{E}_f[\text{acres}] = \frac{b}{F}.$$

A small fraction of farms can control a large fraction of operated acreage. A good farm-count fit need not reproduce land shares closely.

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Entry, exit, and productive-unit turnover

Summing the dynamic balance equations over sizes makes interior transitions cancel:

$$\dot{F} = e - x\mu_1.$$

At stationarity, entrants replace exiting farms, so

$$e = x\mu_1, \qquad \mu_1 = \frac{1-p}{m}.$$

Quantity	Meaning
$1 - p = me/x$	Entrants' share of displaced units
e/F	Entry flow relative to the farm population
$x\mu_1/F$	Exit flow relative to the farm population

Only the first is identified from the fitted ratio alone. Calendar-time entry and exit rates require an intensity scale.

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The entry holding differs from mean farm size

m	Units acquired by each entrant at the moment of entry.
bm	Acreage represented by that entry holding.
n	Current units operated by an incumbent after acquisitions and losses.
$b/F(p,m)$	Mean acreage across all active farms in the stationary distribution.

An estimated entry holding of 66 acres does not imply that the mean farm operates 66 acres.

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Why fitted mean acreage increases with p

Let U be uniform on $\{1, \dots, m\}$ and let independent K satisfy

$$\Pr(K = k) = (1 - p)p^k, \qquad k = 0, 1, \dots.$$

Convolution reproduces the operated-unit distribution:

$$\Pr(U + K = n) = a_n.$$

As p rises, K shifts upward in first-order stochastic dominance. Therefore

$$F(p, m) = \sum_n \frac{a_n}{n} = \mathbb{E}\left[\frac{1}{U + K}\right] \quad \text{strictly decreases.}$$

▸ Ordering check

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The index and model mean acreage

At fixed m and b,

$$\frac{\partial F}{\partial p} < 0, \qquad \frac{\partial (b/F)}{\partial p} > 0, \qquad \frac{\partial p^{-1}}{\partial p} < 0.$$

Hence a standardized p^{-1} is a strictly decreasing transformation of model mean acreage.

- The distribution-based index summarizes farm-size structure.
- It need not be exactly proportional to inverse observed mean acreage.
- Its interpretation as a margin ranking comes from the additional equilibrium restrictions.

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Why stationary counts cannot identify annual rates

The distribution depends on $p = z/x$ and m. For any $c > 0$,

$$(z, x, e) \mapsto (cz, cx, ce)$$

preserves p, $x = z + me$, and the stationary distribution.

- The same cross-section can arise from slow or fast turnover.
- Observing farm-size shares does not supply a calendar clock.
- This is a property of the fixed-rate distributional process. A complete equilibrium also imposes its price and cost conditions.

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The nonstationary balance equation

For fixed hazards,

$$\dot{\mu}_n(t) = z(n-1)\mu_{n-1}(t) + x(n+1)\mu_{n+1}(t) - (z+x)n\mu_n(t) + e\mathbf{1}\{n=m\}.$$

Multiplying all hazards by c multiplies the right-hand side by c. For the same initial distribution,

$$\boldsymbol{\mu}^{(c)}(t) = \boldsymbol{\mu}^{(1)}(ct).$$

The stationary limit is unchanged, while the time scale changes.

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The first moment: a generator calculation

For an incumbent size function $f(n)$, the transition generator is

$$\mathcal{A}f(n) = zn[f(n+1) - f(n)] + xn[f(n-1) - f(n)].$$

For $f(n) = n$,

$$\mathcal{A}n = (z - x)n.$$

Adding entry at size m gives, with $S_1 = \sum_n n\mu_n$,

$$\dot{S}_1 = -(x - z)S_1 + me.$$

If $S_1(0) = 1$ and $me = x - z$, then $S_1(t) = 1$ throughout.

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The second factorial moment

Let $f(n) = n(n-1)$ and $S_2 = \sum_n n(n-1)\mu_n$.

$$f(n+1) - f(n) = 2n, \qquad f(n-1) - f(n) = -2(n-1).$$

Thus

$$\begin{aligned} \mathcal{A}f(n) &= 2zn^2 - 2xn(n-1) \\ &= -2(x-z)n(n-1) + 2zn. \end{aligned}$$

Adding entry yields

$$\dot{S}_2 = -2(x-z)S_2 + 2zS_1 + em(m-1).$$

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Relative adjustment and the half-life

Let $\delta = x - z = me$. With $S_1 = 1$,

$$S_2^* = \frac{2z + em(m-1)}{2\delta}, \qquad S_2(t) - S_2^* = [S_2(0) - S_2^*]e^{-2\delta t}.$$

The half-life of this moment's deviation satisfies

$$T_{1/2} = \frac{\log 2}{2\delta}, \qquad zT_{1/2} = \frac{p \log 2}{2(1-p)}.$$

- p identifies adjustment speed relative to z.
- Converting that speed into years requires z in annual units.

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The interpretation of the two census fits

At the reported median fitted ratios:

	1940	1950
Median p	0.99550	0.99659
Entrant-acquired share $1-p$	0.45%	0.34%
Relative adjustment $2(1-p)/p$	0.00904	0.00684
Dimensionless half-life $zT_{1/2}$	76.7	101.3

These calculations use the rounded reported medians. The two dates are separate descriptive fits within a stationary family. They do not establish attainment of two steady states within ten years.

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Common normalized costs yield a common expansion rate

Impose

$$c_{E,s} = f_E L_s, \qquad c_{I,s} = f_I L_s,$$

with common f_E, f_I, ρ, β_I. Then

$$z_s = \left(\frac{\beta_I f_E}{f_I} \right)^{\beta_I/(1-\beta_I)} \equiv z.$$

Define the common composites

$$d = \rho f_E, \qquad k_{\beta_I} = f_E z > 0.$$

The data need not assign numerical values to these quantities to construct the relative index.

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Deriving the common affine margin mapping

Insert normalized costs into the general margin equation:

$$\begin{aligned}\pi_s &= \frac{f_E\{\rho + z[p_s^{-1} - (1-\beta_I)]\}}{1+\rho f_E} \\ &= \frac{d + k_{\beta_I}[p_s^{-1} - (1-\beta_I)]}{1+d} \\ &= \underbrace{\frac{d - k_{\beta_I}(1-\beta_I)}{1+d}}_{a_\pi} + \underbrace{\frac{k_{\beta_I}}{1+d}}_{b_\pi > 0} p_s^{-1}.\end{aligned}$$

Within the maintained common-cost family, inverse p orders gross operating margins.

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Feasibility and unidentified economic scale

Since $\pi = 1 - \Lambda^{-1}$,

$$\Lambda_s = \frac{1+d}{1 - k_{\beta_I}[p_s^{-1} - (1-\beta_I)]}.$$

Finite positive markups require

$$k_{\beta_I}[p_s^{-1} - (1-\beta_I)] < 1.$$

For any fitted sample, a sufficient common bound is

$$0 < k_{\beta_I} < \frac{1}{\max_{s,\tau} \widehat{p}_{s,\tau}^{-1} - (1-\beta_I)}.$$

This gives a feasible set. It does not select a markup level or estimate cost parameters.

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Constructing the fixed-reference index

Set $h_{s,\tau} = \widehat{p}_{s,\tau}^{-1}$ and use all 48 states in 1940:

$$\bar{h}_{1940} = \frac{1}{48}\sum_{s} h_{s,1940}, \qquad \sigma^2_{h,1940} = \frac{1}{48}\sum_{s}(h_{s,1940} - \bar{h}_{1940})^2.$$

For both dates,

$$\mathcal{M}_{s,\tau} = \frac{h_{s,\tau} - \bar{h}_{1940}}{\sigma_{h,1940}}, \qquad \tau \in \{1940, 1950\}.$$

The denominator uses the population convention 48, not 47.

▸ Index example

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Why the same 1940 scale is used in 1950

- The full 1940 index has mean zero and population standard deviation one.
- The 1950 index uses the same reference mean and scale.
- Its mean and variance need not equal zero and one.
- The 33-state subset retains these same reference moments.
- Separate standardization by date or sample would change the units of comparison.

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Why unknown common costs cancel

Under $\pi_{s,\tau} = a_\pi + b_\pi h_{s,\tau}$ and $b_\pi > 0$,

$$\bar{\pi}_{1940} = a_\pi + b_\pi \bar{h}_{1940}, \qquad \sigma_{\pi,1940} = b_\pi \sigma_{h,1940}.$$

Therefore

$$\frac{\pi_{s,\tau} - \bar{\pi}_{1940}}{\sigma_{\pi,1940}} = \frac{b_\pi (h_{s,\tau} - \bar{h}_{1940})}{b_\pi \sigma_{h,1940}} = \mathcal{M}_{s,\tau}.$$

The common intercept and positive scale cancel exactly.

▸ Invariance check

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The precise relationship to markups

Since $\pi = 1 - \Lambda^{-1}$,

$$\mathcal{M}_{s,\tau} = -\frac{\Lambda_{s,\tau}^{-1} - \overline{\Lambda^{-1}}_{1940}}{\sigma_{\Lambda^{-1},1940}}.$$

Recovering markup levels would require

$$\Lambda_{s,\tau} = \frac{1}{1 - \bar{\pi}_{1940} - \sigma_{\pi,1940}\mathcal{M}_{s,\tau}}.$$

- Higher $\mathcal{M}$ ranks higher conditional margins and markups.
- It equals the negative standardized inverse markup.
- It does not equal the standardized markup or its logarithm.

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General innovation elasticity and index invariance

The stationary family is $\mu_n(p, m)$.

- Neither β_I nor cost parameters enter this family separately.
- Holding data and fitting procedures fixed preserves $\widehat{p}$, $\widehat{m}$, and $\mathcal{M}$.
- A different common admissible β_I changes conditional margin levels and feasibility bounds.
- Its common intercept and scale still cancel under standardization.

This algebraic invariance does not establish robustness to different demand, entry, or preference assumptions.

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Which quantities are identified in this application?

Fitted or constructed	Not separately recovered from farm counts
$\widehat{m}$ within the tested search range	Actual entrant holdings or an annual entry frequency
$\widehat{p}_{s,\tau}$ within the stationary family	Annual levels of z_s, x_s, e_s
$\mathcal{M}_{s,\tau}$ from fitted inverse ratios	Absolute margins, markups, costs, or ρ
Conditional margin ordering	Household entitlement loss or an exploitation rate
Empirical regression associations	A causal effect of margins on subsistence or growth

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The growth implication within the equilibrium family

Let $h = p^{-1}$ and maintain common costs and preferences. Since $x = zh$,

$$g(h) = zh \log \Lambda(h) = zh \log \left(\frac{1+d}{1 - k_{\beta_I}[h - (1-\beta_I)]} \right).$$

Differentiate:

$$g'(h) = z \left[\log \Lambda(h) + \frac{h k_{\beta_I}}{1 - k_{\beta_I}[h - (1-\beta_I)]} \right] > 0.$$

Both terms are positive on the feasible domain $h > 1$, $\Lambda(h) > 1$.

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What varies in the growth comparative static?

- Common costs keep the incumbent rate z fixed.
- Larger $h = x/z$ means more frequent replacement relative to that rate.
- The equilibrium mapping also associates larger h with a larger quality step Λ.
- Both components increase $g = x \log \Lambda$.

This compares equilibria with different quality-step primitives. It does not describe an intervention on an observed profit margin.

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Historical data and the empirical unit

1950 Census of Agriculture, Volume II, Table 11

- Forty-eight historical states in 1940 and 1950.
- Twelve farm-size classes report farm counts and operated acreage.
- Source: 1950 Census of Agriculture, Volume II, *Size of Farm*, Table 11, printed pp. 820–829.
- The source's comparative table includes 1945, but this extension uses 1940 and 1950.
- The national distribution serves as a separate fit benchmark.

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The twelve observed acreage classes

Smaller acreage classes	Larger acreage classes
Under 10	140–179
10–29	180–219
30–49	220–259
50–69	260–499
70–99	500–999
100–139	1,000 and over

The categories describe all land operated by the farm. They do not measure legal ownership, soybean acreage, or productive capacity adjusted for land quality.

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Observed shares and denominator consistency

For reported class counts $N_{s,\tau,k}$,

$$\widehat{f}_{s,\tau,k} = \frac{N_{s,\tau,k}}{\sum_{\ell=1}^{12} N_{s,\tau,\ell}}.$$

- Each distribution uses the total of its own twelve reported classes.
- The 1950 size figures are sample estimates.
- Mixing their denominator with full-census tenure totals would change the shares incorrectly.
- The fitting exercise does not reconstruct the original survey design.

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Mapping model sizes into observed classes

For class interval $[\underline{a}_k, \overline{a}_k)$ and acres per unit b,

$$\mathcal{B}_k(b) = \{n \geq 1 : \underline{a}_k \leq bn < \overline{a}_k\}.$$

Predicted shares are

$$F_k(p, m; b) = \frac{\sum_{n \in \mathcal{B}_k(b)} \mu_n(p, m)}{F(p, m)}, \qquad G_k(p, m; b) = \sum_{n \in \mathcal{B}_k(b)} n\mu_n(p, m).$$

For the top class, $\overline{a}_{12} = \infty$. The complete tail enters its probability, without an arbitrary midpoint.

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Fitting each state for a candidate entry holding

At fixed m and b, minimize cross entropy:

$$\mathcal{L}_{s,\tau}(p, m; b) = -\sum_{k=1}^{12} \widehat{f}_{s,\tau,k} \log F_k(p, m; b).$$

$$\widehat{p}_{s,1940}(m; b) = \arg\min_{0<p<1} \mathcal{L}_{s,1940}(p, m; b).$$

- A zero observed share contributes zero.
- Classes with many observed farms receive more weight within a state.
- The objective penalizes predictions that assign little mass to populated classes.

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Cross entropy and a grouped likelihood

Under a multinomial count interpretation,

$$\log \mathcal{L}^{\text{mult}} = \text{constant} + \sum_k N_k \log F_k.$$

Since $N_k = N \widehat{f}_k$,

$$-\log \mathcal{L}^{\text{mult}} = \text{constant} + N \left[-\sum_k \widehat{f}_k \log F_k \right].$$

The same minimizer follows at fixed N. Cross entropy also works as a descriptive discrepancy criterion without asserting an independent multinomial sampling design.

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The score and local curvature of the grouped fit

Technical detail conditional on the selected m and acreage mapping

Let $\xi = \log[p/(1-p)]$ and write $F_k(\xi)$ for a predicted class probability. The per-farm objective and score are

$$C(\xi) = -\sum_k \widehat{f}_k \log F_k(\xi), \qquad C'(\xi) = -\sum_k \widehat{f}_k \frac{F_k'(\xi)}{F_k(\xi)}.$$

Its second derivative is

$$C''(\xi) = \sum_k \widehat{f}_k \left[\left(\frac{F_k'}{F_k} \right)^2 - \frac{F_k''}{F_k} \right].$$

At an interior optimum the score is zero. Under an independent multinomial working likelihood, local variance is approximately $[NC''(\widehat{\xi})]^{-1}$.

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Selecting the common entry holding

Profile the state fits and give each state equal weight:

$$\widehat{m}(b) = \underset{m \in \{1,\ldots,\lfloor 500/b \rfloor\}}{\arg\min} \frac{1}{48} \sum_{s} \mathcal{L}_{s,1940}\big(\widehat{p}_{s,1940}(m;b), m; b\big).$$

1. Choose a candidate integer m.
2. Fit one p for each 1940 state distribution.
3. Average the 48 minimized discrepancies.
4. Select the entry holding with the smallest profile value.

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Estimated entry holdings across acreage mappings

Acres per unit b	Selected units $\widehat{m}$	Entry acreage $b\widehat{m}$
0.5	136	68
1.0	66	66
5.0	9	45

- These are fitted parameters, not observed holdings of new entrants.
- Changing b changes the discrete support and requires refitting.
- Interior solutions within the tested search range do not establish optimality outside that range.

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Fitting 1950 and constructing both indices

Hold the 1940-selected $\widehat{m}(b)$ fixed:

$$\widehat{p}_{s,\tau}(b) = \arg\min_{0<p<1} \mathcal{L}_{s,\tau}\big(p, \widehat{m}(b); b\big), \quad \tau \in \{1940, 1950\}.$$

Then invert the fitted ratios and standardize using 1940 reference moments.

- Each 1950 ratio fits that state's own 1950 counts.
- Total operated acreage is normalized separately at each date.
- The procedure does not estimate the intervening transition.

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Numerical implementation and checks

Optimization uses

$$\xi = \log \frac{p}{1-p}, \qquad p = \frac{e^{\xi}}{1+e^{\xi}}, \qquad \xi \in [-8, 24].$$

- Entry profiles use grid spacings 0.05 and 0.025, followed by continuous refinement.
- At the selected entry holdings, all 288 state–date profiles have one detected interior grid minimum and positive local curvature.
- Curvature checks at three step sizes differ by at most approximately 4.19×10^{-7} in relative terms.
- Leave-one-state-out entry profiles assess influence.

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Fit discrepancies for counts and acreage

$$TV^F_{s,\tau} = \frac{1}{2}\sum_{k=1}^{12}\left|\widehat{f}_{s,\tau,k} - F_k(\widehat{p}_{s,\tau}, \widehat{m}; b)\right|,$$

$$TV^A_{s,\tau} = \frac{1}{2}\sum_{k=1}^{12}\left|\widehat{g}_{s,\tau,k} - G_k(\widehat{p}_{s,\tau}, \widehat{m}; b)\right|.$$

- Total variation measures probability mass reassigned across classes.
- Count shares enter the fitting criterion.
- Acreage shares assess a different feature, especially the large-farm tail.

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How to read a total-variation discrepancy

Suppose two acreage classes have observed farm shares $(0.50, 0.50)$ and fitted shares $(0.60, 0.40)$. Then

$$TV^F = \frac{1}{2}\left(|0.50 - 0.60| + |0.50 - 0.40|\right) = 0.10.$$

- Ten percent of farm-count probability mass must move between classes to reconcile the distributions.
- It is not a ten-percent error in total farm numbers: both share distributions sum to one.
- A separate land-share discrepancy gives greater weight to holdings containing many acres.

The estimation criterion fits *farm counts*. A favorable count fit does not guarantee a favorable acreage fit, especially in the open-ended top class.

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Reported baseline distribution fit

State summary	1940	1950
Median fitted p	0.99550	0.99659
Median farm-share TV	15.95%	15.60%
Maximum farm-share TV	46.87%	44.68%
Median land-share TV	18.15%	17.51%

The fitted family summarizes historical structure unevenly across states. A visually reasonable national fit does not establish a uniformly good state fit.

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National farm-count fit: 1940



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National farm-count fit: 1950



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The subsistence index comes from a separate model

The earlier structural-transformation model implies

$$\ell_{a,s,t} = \alpha + (1-\alpha)\frac{\bar{a}_s}{A_{a,s,t}}.$$

Conditional on α, early observations recover

$$\bar{a}_s(\alpha) = \frac{1}{|T_E|}\sum_{t\in T_E} A_{a,s,t}\frac{\ell^{\text{obs}}_{a,s,t} - \alpha}{1-\alpha}, \qquad T_E = \{1960, 1970\}.$$

The common α minimizes full-panel squared prediction error. The farm-dynamics regressions use the resulting fixed $\log \widehat{\bar{a}}_s$.

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The observed productivity-growth outcome

The empirical growth variable is

$$G_s^A = \frac{100}{50}\left[\log A_{a,s,2010} - \log A_{a,s,1960}\right].$$

Here A_a is the ERS agricultural output quantity index divided by the labor input quantity index.

- G_s^A measures annualized log labor-productivity growth.
- Its unit is percent per year in the continuous-growth convention.
- It differs from TFP growth and from model quality growth g_s.
- Both historical indices are related to this same 1960–2010 outcome.

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The two empirical specifications

For each historical date τ separately,

$$\log \widehat{\bar{a}}_s = \beta_{0,\tau} + \beta_{1,\tau}\mathcal{M}_{s,\tau} + \delta_{r(s),\tau} + \varepsilon_{s,\tau},$$
$$G^A_s = \gamma_{0,\tau} + \gamma_{1,\tau}\mathcal{M}_{s,\tau} + \psi_{r(s),\tau} + u_{s,\tau}.$$

- Region effects refer to the four Census regions.
- Unadjusted versions omit region effects.
- The regressions are cross-sectional, not state fixed-effects panels.
- The farm model does not impose the sign of $\beta_{1,\tau}$.

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The full and high-match samples

Full sample	48 states.
High-match sample	Panel A only: 33 states whose subsistence-trajectory scores are strictly above 95. Panel B retains all 48 states.
What selection measures	Similarity of standardized structural-transformation trajectories.
What stays fixed	Estimated m, state ratios, and all-48-state 1940 index moments.

The high-match criterion measures neither farm-size fit nor parameter precision. It does not mean the index is recalibrated for 33 states.

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Estimation and reported uncertainty

- All farm-index association regressions use unweighted OLS.
- HC3 standard errors account for heteroskedasticity and observation leverage.
- Student-t inference uses the regression's residual degrees of freedom.
- Specifications exclude the tenure PCA score. Growth models also omit initial productivity.
- Inference conditions on generated indices, calibrated outcomes, and sample selection.
- Significance levels have no multiple-comparison adjustment.

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Subsistence: full sample

Farm-size date	1940	1940	1950	1950
Region effects	No	Yes	No	Yes
$\mathcal{M}$ coefficient	-0.701^{***}	-0.628^{***}	-0.873^{***}	-0.767^{***}
HC3 standard error	(0.181)	(0.207)	(0.243)	(0.267)
Observations	48	48	48	48
R^2	0.346	0.599	0.365	0.615

HC3 standard errors; Student-t tests. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

- All four coefficients are negative and significant at 1% under the reported inference.
- Higher historical $\mathcal{M}$ is associated with lower calibrated subsistence.

▸ Coefficient calculation

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Subsistence: high-match sample

Farm-size date	1940	1940	1950	1950
Region effects	No	Yes	No	Yes
$\mathcal{M}$ coefficient	-0.414^{***}	-0.463^{*}	-0.512^{**}	-0.562^{*}
HC3 standard error	(0.145)	(0.255)	(0.193)	(0.312)
Observations	33	33	33	33
R^2	0.248	0.362	0.258	0.380

HC3 standard errors; Student-t tests. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

- Signs remain negative under both dates and both control choices.
- The region-adjusted estimates are significant at 10%, but not at 5%.
- The same index units apply as in the full sample.

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How the within-region graphs relate to the regression

Let $x_s = \mathcal{M}_{s,\tau}$ and $y_s = \log \widehat{a}_s$. Subtract means within Census region and within the selected sample:

$$\widetilde{x}_s = x_s - \bar{x}_{r(s)}, \qquad \widetilde{y}_s = y_s - \bar{y}_{r(s)}.$$

The region-adjusted slope is

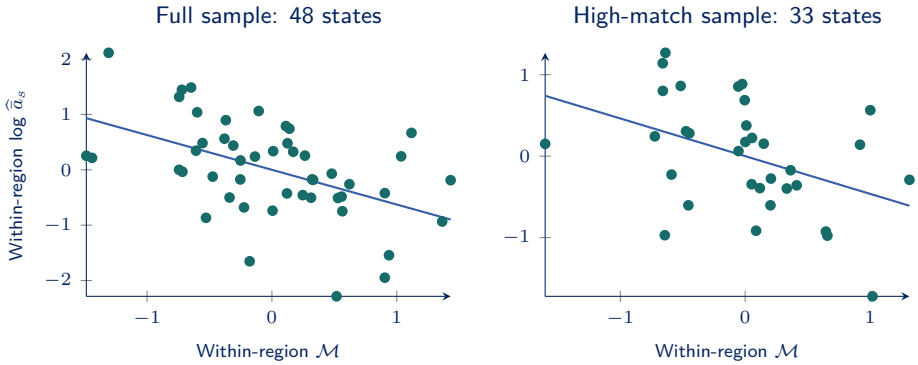
$$\widehat{\beta}_1 = \frac{\sum_s \widetilde{x}_s \widetilde{y}_s}{\sum_s \widetilde{x}_s^2}.$$

The plots display these residuals and the fitted slope. They remove regional means, rather than adding a different model to the table.

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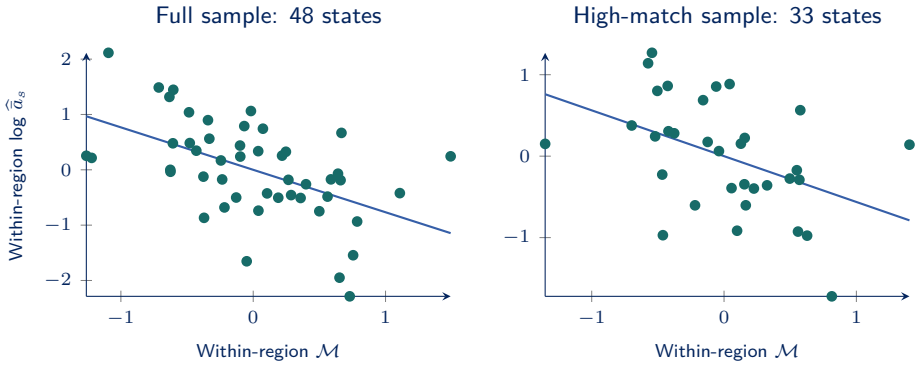
Within-region subsistence association: 1940



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Within-region subsistence association: 1950



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Productivity growth: full sample

Farm-size date	1940	1940	1950	1950
Region effects	No	Yes	No	Yes
$\mathcal{M}$ coefficient	0.415^{**}	0.327^{*}	0.528^{**}	0.442^{*}
HC3 standard error	(0.185)	(0.177)	(0.237)	(0.246)
Observations	48	48	48	48
R^2	0.118	0.453	0.130	0.466

HC3 standard errors; Student-t tests. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

- All signs are positive.
- Region-adjusted estimates are significant at 10%, but not at 5%.
- Their units are percentage points of annualized log growth per index unit.

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Why the growth comparison uses all 48 states

- Table 5, Panel A reports subsistence associations in both the full and high-match samples.
- Panel B reports productivity-growth associations only for the full 48-state sample, for 1940 and 1950, with and without region effects.
- The high-match criterion assesses the separate subsistence calibration. It is not a quality screen for observed productivity growth or farm-size fitting.
- Both panels retain the same 1940 index reference moments and the same fitted historical ratios.

The latest table has two panels: subsistence and observed growth. It does not report a suitability outcome regression.

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The empirical growth comparison

The theory orders $g_s = x_s \log \Lambda_s$ within a restricted equilibrium family. The regressions examine observed G_s^A.

- Their positive signs align with the theoretical direction.
- They do not identify annual hazards or the quality-step parameter.
- ERS output per labor input is not the same object as Q_s.
- Initial-productivity convergence remains a competing explanation because the models omit initial productivity.

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Productivity and the recovered subsistence index

For one early-year recovery moment,

$$\bar{a}_{s,t}(\alpha) = A_{a,s,t}\frac{\ell^{\text{obs}}_{a,s,t} - \alpha}{1-\alpha}.$$

Holding α fixed,

$$\frac{\partial \bar{a}_{s,t}}{\partial A_{a,s,t}} = \frac{\ell^{\text{obs}}_{a,s,t} - \alpha}{1-\alpha} > 0, \qquad \frac{\partial \bar{a}_{s,t}}{\partial \ell^{\text{obs}}_{a,s,t}} = \frac{A_{a,s,t}}{1-\alpha} > 0,$$

when the labor-share gap is positive. Higher productivity and lower agricultural labor absorption enter this recovery formula in opposite directions.

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The paper's conditional accounting interpretation

- Productive reorganization may generate productivity gains and reduce agricultural labor absorption together.
- At a fixed positive labor-share gap, higher early productivity raises recovered subsistence.
- This component can offset the lower index associated with labor displacement.
- Later growth associations do not establish the source of early productivity levels or determine the net direction of endogeneity bias.

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Historical interpretation and primitive accumulation

- Replacement transfers an operating position while raising unit productivity.
- This mechanism separates productive advance from incumbent survival.
- The empirical index is associated with lower subsistence and, less precisely, faster subsequent productivity growth.
- A Marxian interpretation asks how productive improvements accompany the reorganization of claims to resources and livelihoods.

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Thin description and the interpretation of outcomes

The observed outcomes provide partial views of a historical process.

- Productivity gains can coexist with changes in access to agricultural livelihoods.
- Greater output per worker does not, by itself, describe who retains productive claims or receives the gains.
- Farm-size organization, subsistence, and growth are different analytical views, with shared inputs and limitations.
- Their combination motivates an account of transformation beyond efficient labor reallocation alone.

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Boundaries of the model's explanation

Model feature	Remaining empirical distinction
Transfer of operating control	Purchase, rental, coercion, exclusion, and compensation are not observed.
Conditional operating margin	Actual profits and rent extraction require additional cost and income information.
Representative household	The model does not allocate gains and losses among displaced families and asset owners.
Stationary size family	Census fits do not estimate a historical transition.

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The complete logical chain

1. Demand and limit pricing make profit per operated unit common.
2. Incumbent optimization and entry determine values and expansion.
3. Resource clearing links margins to the expansion/displacement ratio.
4. Stationary flows map that ratio into farm-size probabilities.
5. Historical counts fit the ratio and common entry holding.
6. Standardization supplies $\mathcal{M}$ without numerical cost anchors.
7. Regressions connect this historical summary to later subsistence and growth.

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From differential surplus to the margin index

At the limit price, the competing technology just covers production costs:

$$\underbrace{\Pi^I_{sj}}_{\text{surplus flow}} = \underbrace{(MC^{old}_{sj} - MC_{sj})}_{\text{cost advantage}} \underbrace{y_{sj}}_{\text{output}} = \pi_s Y_s.$$

Its revenue share and standardized comparison are different quantities:

$$\underbrace{\pi_s}_{\text{surplus share}} = \frac{\Pi^I_{sj}}{P_{sj} y_{sj}} = 1 - \Lambda_s^{-1}, \qquad \underbrace{\mathcal{M}_{s,\tau}}_{\text{relative index}} = \frac{\pi_{s,\tau} - \bar{\pi}_{1940}}{\mathrm{sd}_{1940}(\pi)}.$$

$$\pi_{s,\tau} - \bar{\pi}_{1940} = \underbrace{b_\pi \, \mathrm{sd}_{1940}(\widehat{p}^{-1})}_{\text{unidentified positive scale}} \mathcal{M}_{s,\tau}.$$

One index unit is one reference standard deviation, not one percentage point of profit. Surplus income also depends on revenue.

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Why fixed-resource reallocation helps identify relative margins

1. **Conservation:** $x = z + me$ ties expansion and entry to displacement.
2. **Distribution:** dividing stationary balances by x replaces the entry term with $(1-p)/m$. Farm-count shares fit p and m.
3. **Equilibrium:** conservation cancels the rate term in labor clearing; log utility supplies the common growth-adjusted discount rate.
4. **Comparison:** common normalized costs imply $\pi = a_\pi + b_\pi p^{-1}$ with $b_\pi > 0$. Standardization cancels the unknown intercept and slope.

The substantive restriction and the inference strategy meet at the same point: acquiring an operating position means reallocating an existing claim to production and its surplus.

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What the farm extension contributes

- A tractable model of expansion, displacement, entry, and endogenous exit.
- A fitted summary of historical operated farm-size distributions.
- A transparent conditional bridge from size structure to relative operating margins.
- A growth ordering under explicitly stated common equilibrium restrictions.
- Descriptive links to the paper's broader question about provisioning relations.

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§5

Cash Crops and SNAP

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State SNAP participation

$$y_s = \log\left(\frac{\text{SNAP persons}_{s,\,Feb\ 2026}}{\text{Population}_{s,\,July\ 1\ 2025}}\right).$$

- Administrative participants come from USDA Food and Nutrition Service records.
- The population denominator comes from the Census Vintage 2025 estimates.
- The outcome measures persons, while the county outcome measures household receipt.
- SNAP reflects need, eligibility, administrative access, and take-up. It does not directly measure hunger.

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Eligibility and take-up are different margins

For illustration, suppose 1,000 people are in a population, 200 are eligible for assistance, and 150 eligible people participate.

$$\frac{\text{participants}}{\text{population}} = \frac{\text{eligible}}{\text{population}} \times \frac{\text{participants}}{\text{eligible}} = 0.20 \times 0.75 = 0.15.$$

- The eligibility share reflects income, household circumstances, and program rules.
- Take-up among eligible people also reflects information, access, and administrative burdens.
- A higher participation rate can therefore mean greater need, better access, or both.

This is an illustrative decomposition assuming participants belong to the eligible group. The paper does not separately estimate these two components.

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Historical state crop intensity

First average acreage over the historical period:

$$\overline{\text{Acres}}_{c,s} = \frac{1}{T} \sum_{t=1926}^{1940} \text{Acres}_{c,s,t}.$$

Then construct the excluded instrument

$$Z_{c,s} = \log\left(1 + 1000\,\frac{\overline{\text{Acres}}_{c,s}}{\text{LandAcres}_s}\right).$$

The acreage-intensity unit is acres per 1,000 state land acres. Land acres come from county Gazetteer areas aggregated to the state.

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Intensity and crop share are different variables

The interaction specifications use

$$\text{Share}_{c,s} = \frac{\overline{\text{Acres}}_{c,s}}{\overline{\text{Acres}}_{cotton,s} + \overline{\text{Acres}}_{corn,s} + \overline{\text{Acres}}_{soybean,s}}.$$

- Intensity measures historical exposure relative to all state land.
- Crop share measures composition among the three selected crops.
- A soybean share is not the soybean fraction of all state acres or all crop acres.
- The IV and interaction designs use these different variables for different questions.

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Why the log-plus-one convention must remain explicit

Chen and Roth (2024)

- $\log(1+x)$ retains a reported zero and compresses large exposure values.
- The expression is not an ordinary logarithm of strictly positive acreage.
- A coefficient on it is not automatically an elasticity.
- Changing the units inside the transformation can change the fitted relationship.
- The paper fixes acres per 1,000 land acres rather than varying units to obtain a preferred result.

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Fieldwork suitability and geographical controls

- Suitability is mean days suitable for fieldwork across available April–October weekly observations, 1995–2026.
- It summarizes a broad agricultural condition, not a direct measure of historical entitlement.
- Census-region effects distinguish Northeast, Midwest, South, and West.
- Within-region estimation removes differences in regional intercepts but not all historical or institutional confounding.
- Redundant indicators are omitted when the sample has no independent variation for a category.

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The baseline subsistence–SNAP relationship

$$y_s = b_0 + \beta \log \widehat{a}_s + \gamma_{r(s)} + \varepsilon_s.$$

- β is a log-log association between the calibrated index and the SNAP participation rate.
- Region effects compare states within broad regions.
- OLS and fit-weighted versions examine sensitivity to weighting.
- The preferred sample contains 33 states with trajectory score above 95.
- The index remains generated by the separate structural calibration.

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What "controlling for region" means algebraically

Let D contain an intercept and the nonredundant region indicators. Define

$$M_D = I - D(D'D)^{-1}D', \qquad \widetilde{y} = M_D y, \qquad \widetilde{a} = M_D a.$$

For unweighted OLS, the Frisch–Waugh–Lovell identity gives

$$\widehat{\beta} = \frac{\widetilde{a}'\widetilde{y}}{\widetilde{a}'\widetilde{a}}.$$

With only region indicators, residualization subtracts the corresponding region mean. The slope therefore compares deviations from regional averages.

- It does not remove every difference between states in the same region.
- Weighted regressions require the corresponding weighted projection, not ordinary region demeaning.

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What the two IV stages do

Let $a_s = \log \widehat{\bar{a}}_s$. The first stage is

$$a_s = \pi_0 + \pi_1 Z_{soybean,s} + C_s'\theta + \gamma_{r(s)} + u_s.$$

The second-stage representation is

$$y_s = b_0 + \beta \widehat{a}_s + C_s'\lambda + \gamma_{r(s)} + \varepsilon_s.$$

- Historical soybean intensity selects a predicted component of calibrated subsistence.
- 2SLS relates that component to contemporary SNAP participation.
- C_s is empty in Panel A, tenure PCA in B, and tenure PCA plus suitability in C. All panels retain region effects.
- Valid IV uncertainty uses the IV covariance calculation, not naive OLS standard errors from fitted values.

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The two-stage estimator in one expression

Technical detail: the same controls must appear in both stages

Let $X = [D, a]$ contain the controls and calibrated log subsistence, and let $Z = [D, Z_{soybean}]$ contain controls and the excluded instrument. Then

$$P_Z = Z(Z'Z)^{-1}Z', \qquad \widehat{b}_{2SLS} = (X'P_Z X)^{-1}X'P_Z y.$$

1. First project the regressors onto the instrument space: $\widehat{X} = P_Z X$.
2. Regress y on $\widehat{X}$ to obtain the same coefficient estimates.
3. For inference, use the structural residual $\widehat{\varepsilon} = y - X\widehat{b}$, together with the IV covariance formula.

For positive observation weights, transform y, X, and Z by $W^{1/2}$ first. Naive second-stage OLS uncertainty ignores the IV construction.

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Relevance and exclusion answer different questions

Relevance	Does historical soybean intensity predict $\log \widehat{\bar{a}}_s$ after the controls? The first stage addresses this question.
Exclusion	Does the instrument affect contemporary SNAP only through calibrated subsistence, conditional on the controls? A strong first stage cannot establish this restriction.
Other pathways	Land institutions, settlement, racial exclusion, credit, and market integration may connect historical specialization to current SNAP through omitted channels.

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The just-identified ratio interpretation

After residualizing variables on the same controls, write the reduced form as

$$\widetilde{y}_s = \delta \widetilde{Z}_s + v_s, \qquad \widetilde{a}_s = \pi_1 \widetilde{Z}_s + u_s.$$

With one excluded instrument and one endogenous regressor,

$$\widehat{\beta}_{IV} = \frac{\widehat{\delta}}{\widehat{\pi}_1}.$$

- A negative first stage and a positive reduced form produce a negative IV ratio.
- A small first-stage denominator makes that ratio imprecise.
- Historical timing alone does not validate exclusion.

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Baseline first-stage results

Outcome: $\log \widehat{a}_s$	Unweighted	Fit-weighted
Soybean log-plus-one intensity	-0.409^{***}	-0.408^{***}
Standard error	(0.104)	(0.104)
First-stage F	15.42	15.56
Region effects	Yes	Yes
States	33	33

Reported heteroskedasticity-robust inference; $^{***}p < 0.01$.

Greater historical soybean exposure predicts lower calibrated subsistence in this sample. The reported F-statistics address relevance rather than exclusion.

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Baseline SNAP estimates

Outcome: log SNAP rate	OLS	Soybean IV	Soybean IV-WLS
$\log \widehat{a}_s$	-0.161^{***}	-0.195^{**}	-0.196^{**}
Standard error	(0.050)	(0.083)	(0.083)
Region effects	Yes	Yes	Yes
Fit weights	No	No	Match/100
States	33	33	33

Reported heteroskedasticity-robust inference; $^{**}p < 0.05$, $^{***}p < 0.01$.

The signs agree across these specifications. Regression-stage bootstrap intervals, discussed below, show substantial finite-sample uncertainty.

▸ Log-log calculation

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First stage across control sets

Table 6, columns (2)–(3): outcome is $\log \widehat{a}_s$

	A: Baseline	B: Tenure	C: Tenure + suitability
Soybean intensity, unweighted	-0.409^{***}	-0.276^{***}	-0.277^{**}
Robust SE	(0.104)	(0.103)	(0.129)
Soybean intensity, WLS	-0.408^{***}	-0.275^{***}	-0.278^{**}
Robust SE	(0.104)	(0.103)	(0.129)
First-stage F, unweighted	15.42	7.16	4.59
First-stage F, WLS	15.56	7.18	4.66
Region effects	Yes	Yes	Yes
States	33	33	33

- **A to B:** the coefficient becomes smaller in magnitude; its SE barely changes.
- **B to C:** the coefficient barely changes; its SE increases.

WLS weights: match score/100. $^{**}p < 0.05$, $^{***}p < 0.01$.

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SNAP estimates across the same control sequence

Table 6, columns (1), (4), and (5): coefficient on $\log \widehat{a}_s$

	A: Baseline	B: Tenure	C: Tenure + suitability
OLS	-0.161^{***}	-0.202^{***}	-0.184^{***}
Robust SE	(0.050)	(0.064)	(0.071)
Soybean IV	-0.195^{**}	-0.276^{*}	-0.181
Robust SE	(0.083)	(0.145)	(0.189)
Soybean IV-WLS	-0.196^{**}	-0.275^{*}	-0.180
Robust SE	(0.083)	(0.145)	(0.189)
First-stage F, IV / IV-WLS	15.42 / 15.56	7.16 / 7.18	4.59 / 4.66
Region effects	Yes	Yes	Yes
States	33	33	33

Negative IV signs persist while uncertainty increases. The coefficient does *not* move monotonically toward zero. OLS remains significant at 1% throughout.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. WLS: match score/100.

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What the first-stage F comparison tells us

The reported first-stage test asks whether soybean intensity adds predictive information for subsistence *conditional on the included controls*.

- $15.42 \rightarrow 7.16 \rightarrow 4.59$ describes declining conditional first-stage strength, not a sequence of p-values.
- Tenure adjustment removes part of the soybean–subsistence association: $-0.409 \rightarrow -0.276$.
- Suitability adds little further attenuation: $-0.276 \rightarrow -0.277$, but the SE rises from 0.103 to 0.129.
- The comparison concerns the same 33 states and region effects, so the changes do not arise from changing the estimation sample.

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Crop comparisons: baseline region-adjusted estimates

Instrument	First-stage coefficient	First-stage F	IV coefficient
Cotton intensity	0.082 (0.075)	1.18	0.150 (0.313)
Corn intensity	−0.172 (0.153)	1.28	−0.420 (0.336)
Soybean intensity	−0.409 (0.104)	15.42	−0.195 (0.083)

33 states; unweighted estimation with region effects. Standard errors in parentheses.

- Soybean intensity supplies the strongest reported first stage among these choices.
- Suitability is now an included control in the soybean specification, not an excluded instrument.
- A negative corn estimate without precision does not establish an equivalent effect.

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Appendix crop comparisons after tenure adjustment

Instrument / included control	First stage (SE)	F	IV (SE)
Cotton + tenure	-0.138 (0.067)	4.17	-0.291 (0.217)
Corn + tenure	-0.283 (0.120)	5.53	-0.299 (0.154)
Soybean + tenure	-0.276 (0.103)	7.16	-0.276 (0.145)
Soybean + suitability	-0.433 (0.117)	13.75	-0.130 (0.115)
Soybean + both	-0.277 (0.129)	4.59	-0.181 (0.189)
Unweighted; region effects throughout; the same 33 states.			

- The first three rows compare crop instruments with tenure held constant.
- The final two rows use soybean as the excluded instrument; suitability is included in both stages.
- Tables E.1–E.3 also report match-score and squared-score weighting. Their left/right blocks now compare absence/presence of tenure, not absence/presence of region effects.

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Tenure and historical land access

- The 1940 PCA index is oriented toward greater tenancy and less full ownership.
- Its association with log subsistence, holding Census divisions constant, is 0.434^{***} in the full sample and 0.388^{**} in the high-match sample (Table 3).
- Controlling for that composition reduces the soybean first-stage magnitude from 0.409 to 0.276 (Table 6).

Together, these comparisons locate part of the soybean–subsistence relationship within the historical organization of land access. The score measures tenure composition, not the strength of individual rights: higher tenancy is not automatically stronger entitlement.

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Fieldwork conditions and operating capacity

- Cultivation requires planting, field operations, and harvesting within available working periods.
- For a fixed acreage and task requirement, fewer workable days require greater daily capacity to complete those tasks.
- The consequence depends on command over machinery, labor, and land improvements, not simply the area legally available to cultivate.

Apland (1993) shows that field-day variability affects profitability, risk, and economical machinery requirements; expanding acreage with fixed machinery and labor raises exposure to delays.

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Historical drainage: institutions and investment

Edwards and Thurman (2025)

- Eastern U.S. drainage frequently required connected improvements across separately owned parcels.
- State enabling legislation adopted between 1859 and 1912 supported drainage districts with taxing, financing, and eminent-domain powers.
- Landowners retained farm operation while these institutions enabled collective improvements.
- Improved acreage and land values increased in naturally wet counties relative to drier counties within the same states.

The historical connection is specific: productive workability depended on an organization of rights and investment, not on natural conditions or technology alone.

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Mechanization, timeliness, and livelihoods

Olmstead and Rhode (2001), American agriculture, 1910–1960

- Timeliness and seasonal capacity constraints contributed to tractor adoption.
- Farm scale and tractor adoption were positively related in both directions.
- Tractors reduced agricultural labor requirements and enabled timely operations with less reliance on hired labor.

An adopting operator could gain operational autonomy while production required fewer workers. Higher productivity, greater acreage capacity, and access to agricultural employment therefore need not expand together.

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Contemporary contracts and investment incentives

Chakravorty et al. (2024)

In a 2024 Corn Belt survey, 97 of the 463 respondents answering the relevant question, about 21%, identified rental-contract specifications as a substantial barrier to drainage investment.

- Access to cultivate and sufficiently durable access to recover an improvement's returns are different claims.
- Contract duration and compensation arrangements can affect whether a tenant can justify the investment.
- This is a contemporary illustration; the drainage and tractor studies supply the historical evidence.

The comparison concerns command over productive services, not fixed farm-size categories. Equipment rental and sharing can also serve smaller farms (MacDonald et al. 2013).

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Primitive accumulation: the control comparison

1. The significant baseline IV relationship connects historical soybean specialization, lower calibrated subsistence, and contemporary food assistance.
2. The tenure–subsistence association and first-stage attenuation place part of that relationship within historical land-access arrangements.
3. Fieldwork conditions add an operational dimension: productive participation depends on timely services, investment, and the ability to retain returns.

Although these comparisons do not directly trace dispossession within the sample, historical evidence documents how rights, investment, and mechanization reorganized productive participation. Together they support a reading of commercialization as a reorganization of access to production and subsistence.

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The interaction specification is separate from IV

$$y_s = b_0 + \beta_1 \log \widehat{\bar{a}}_s + \beta_2 \text{Share}_{c,s} + \beta_3 (\log \widehat{\bar{a}}_s \times \text{Share}_{c,s}) + C_s' \lambda + \gamma_{r(s)} + \varepsilon_s.$$

Each specification uses one crop share. Tenure and suitability, when included in C_s, are not interacted.

$$\frac{\partial \widehat{y}_s}{\partial \log \widehat{\bar{a}}_s} = \beta_1 + \beta_3 \text{Share}_{c,s}.$$

The interaction asks whether the subsistence–SNAP slope differs across historical agricultural conditions. It does not instrument both interacted regressors.

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The baseline soybean interaction

Fit-weighted regression with region effects	Estimate (SE)
$\log \widehat{a}_s$	-0.230^{***} (0.082)
Soybean share	11.968^{*} (6.755)
$\log \widehat{a}_s \times$ soybean share	2.622^{*} (1.494)
States	33

$^{*}p < 0.10$, $^{***}p < 0.01$. Crop-share main effects are conditional on the interacted subsistence index.

$$\text{Fitted subsistence slope} = -0.230 + 2.622\text{Share}_{soybean,s}.$$

Higher soybean shares make this fitted slope less negative. The main effect -0.230 refers to zero soybean share, and 11.968 refers to $\log \widehat{a}_s = 0$.

▸ Interaction calculation

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Soybean interactions across control sets

Table 6, column (6): match-score weighted, with region effects

	A: Baseline	B: Tenure	C: Tenure + suitability
$\log \widehat{a}_s$	-0.230^{***}	-0.274^{***}	-0.259^{**}
HC1 SE	(0.082)	(0.090)	(0.106)
Soybean share	11.968^{*}	13.301^{**}	11.460
HC1 SE	(6.755)	(6.735)	(8.309)
$\log \widehat{a}_s \times$ soybean share	2.622^{*}	2.852^{*}	2.506
HC1 SE	(1.494)	(1.483)	(1.783)
Region effects	Yes	Yes	Yes
States	33	33	33

The interaction stays positive. It is significant at 10% in A and B, but not C. Tenure and suitability enter additively; neither is the interacted variable.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. These are descriptive regressions, not IV.

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What fit weighting changes

$$\widehat{\theta}_{WLS} = (X'WX)^{-1}X'Wy, \quad W = \mathrm{diag}(w_s).$$

- Weighting changes how strongly each state influences the fitted relationship.
- $w_s = \mathrm{Match}_s/100$ and its square favor closer trajectory matches.
- Similar OLS and WLS estimates demonstrate limited sensitivity to these particular weights.
- They do not show that the weights equal inverse sampling variances or remove selection concerns.

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What a standard error measures here

- A coefficient is the fitted association in the stated sample and specification.
- Its standard error describes uncertainty under the covariance estimator and reference distribution used by that table.
- A confidence interval combines the coefficient, its standard error, and a critical value; stars summarize thresholds for a specified null hypothesis.
- Statistical significance does not establish a large economic effect, correct model assumptions, or a causal mechanism.

The paper uses different inference conventions across exercises. The farm associations use HC3 with Student-t inference; the tenure tables and county table use their explicitly reported alternatives.

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HC3: the leverage-adjusted covariance calculation

Technical detail for the unweighted farm-association regressions

With design row x_s', OLS residual $\widehat{u}_s$, and leverage

$$h_{ss} = x_s'(X'X)^{-1}x_s,$$

the HC3 covariance is

$$\widehat{\mathrm{Var}}_{HC3}(\widehat{b}) = (X'X)^{-1}\left[\sum_s \frac{\widehat{u}_s^2}{(1-h_{ss})^2}x_s x_s'\right](X'X)^{-1}.$$

High-leverage observations receive larger residual adjustments. The lecture's farm-table standard errors use this covariance and the reported Student-t reference.

- Heteroskedasticity-robust covariance is not the same as correction for estimated regressors.
- It also does not reconstruct uncertainty in the subsistence calibration or in the stationary distributional specification.

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Threshold sensitivity

Threshold	Weighting	IV coefficient (SE)	First-stage F	States
> 85	None	-0.166 (0.110)	6.36	40
	Match/100	-0.167 (0.109)	6.80	40
	Squared	-0.167 (0.107)	7.26	40
> 90	None	-0.149 (0.106)	6.87	38
	Match/100	-0.151 (0.105)	7.31	38
	Squared	-0.153 (0.104)	7.74	38
> 95	None	-0.195^{**} (0.083)	15.42	33
	Match/100	-0.196^{**} (0.083)	15.56	33
	Squared	-0.196^{**} (0.084)	15.69	33

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The regression-stage bootstrap

1. Start with the relevant threshold sample of states.
2. Draw states with replacement, retaining the state-level variables together.
3. Re-estimate the IV specification on that draw.
4. Repeat 999 times and summarize the resulting coefficient distribution.

This exercise holds previously calibrated quantities fixed. It does not re-estimate the full structural calibration in each draw.

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Bootstrap intervals and conventional significance

Weighting	IV point estimate	Bootstrap 95% interval
Unweighted	-0.195	$[-0.415,\ 0.016]$
Match/100	-0.196	$[-0.445,\ -0.003]$
Squared weight	-0.196	$[-0.414,\ -0.007]$

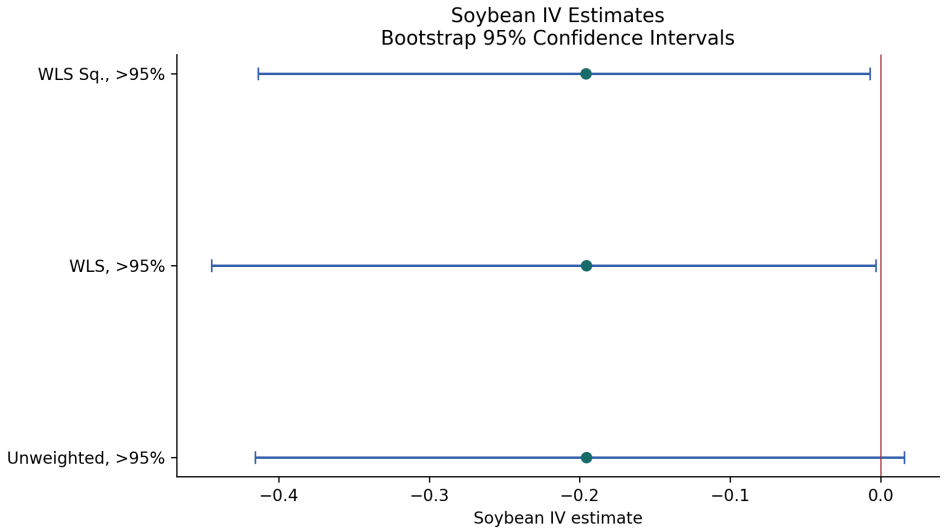
Intervals condition on the precomputed calibration. The unweighted interval includes zero.

- Both weighted intervals exclude zero; the unweighted interval slightly includes it.
- Weighting preserves point estimates close to -0.20, but does not uniformly narrow the intervals.
- The agreement across weighting choices supports the baseline relationship's empirical stability within this 33-state sample.
- Conventional robust stars and these bootstrap intervals summarize different inference procedures.

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The bootstrap figure



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Allowing a bounded direct instrument association

Motivated by Conley et al. (2012)

Suppose the outcome equation also contains γZ_s:

$$y_s = \beta a_s + \gamma Z_s + \text{controls} + \varepsilon_s.$$

With first-stage slope π_1 and reduced-form slope δ,

$$\delta = \beta\pi_1 + \gamma, \qquad \beta(\gamma) = \frac{\delta - \gamma}{\pi_1}.$$

The reported sensitivity allows a direct effect bounded at 10%, 25%, or 50% of the observed reduced-form association. These are assumptions about departures from exclusion.

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Plausibly exogenous point-estimate ranges

Weighting	10% bound	25% bound	50% bound
None	$[-0.215, -0.176]$	$[-0.244, -0.147]$	$[-0.293, -0.098]$
Match/100	$[-0.215, -0.176]$	$[-0.245, -0.147]$	$[-0.294, -0.098]$
Squared	$[-0.215, -0.176]$	$[-0.245, -0.147]$	$[-0.294, -0.098]$

- Sign preservation is mechanical for bounds below 100% of the reduced form; it is not an additional empirical test.
- These ranges are not confidence intervals.
- They do not establish that the permitted bounds cover all historically plausible violations.

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The direct-effect threshold, made explicit

Using the same controls and weights, let $\widehat{\delta}$ be the reduced form and $\widehat{\kappa}$ the first stage:

$$\widehat{\beta}(\eta) = \frac{\widehat{\delta} - \eta}{\widehat{\kappa}}, \qquad |\eta| \le c|\widehat{\delta}|.$$

For the negative baseline IV estimate, the allowed range is

$$\left[(1+c)\widehat{\beta}_{IV},\ (1-c)\widehat{\beta}_{IV}\right].$$

- If $c < 1$, both endpoints retain the negative baseline sign by construction.
- The point estimate reaches zero at $\eta = \widehat{\delta}$: the entire reduced-form association is assigned to the direct effect.
- These are sensitivity scenarios, not confidence intervals or estimated bounds on the historical direct effect.

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A county extension with a different outcome unit

$$\log(\text{SNAPRate}_{c,s}) = \alpha_s + \beta \log\left(1 + 1000\frac{\text{SoybeanAcres}_{c,1919}}{\text{LandAcres}_{c,2024}}\right) + X'_{c,s}\Gamma + \varepsilon_{c,s}.$$

- Outcome: ACS 2024 five-year household SNAP receipt rate.
- Historical exposure: 1919 soybean acres from NHGIS agricultural tables.
- State effects restrict the comparison to counties within states.
- This is a reduced-form extension, not county-level subsistence calibration or county 2SLS.

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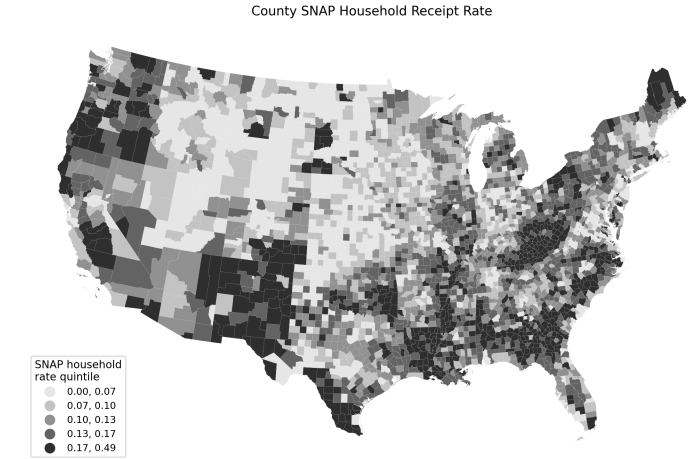
County data coverage and missing values

- The reported common sample contains 562 counties and county equivalents in eight state-level clusters.
- Reported zero soybean acreage remains in the data.
- Missing or invalid required source values are excluded rather than interpreted as zero.
- Identifier matching does not harmonize historical county boundary changes.
- Contemporary SNAP map coverage is broader than the regression's historical-data coverage.

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County SNAP household receipt



Notes: Mainland U.S. counties only; Alaska and Hawaii omitted for readability. Counties grouped into five quantile bins. Missing counties are white with diagonal hatching. N = 3109.

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The three county control groups

Population	Log population.
Demographics	Black population share, Hispanic population share, and share with a bachelor's degree or higher.
Socioeconomic conditions	Log median household income, poverty rate, and unemployment rate.

All eight specifications include state effects and use the same 562 observations. Changes across columns therefore reflect conditioning rather than sample changes.

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County estimates: columns 1–4

	(1)	(2)	(3)	(4)
Soybean intensity	0.024	0.003	-0.056^{***}	0.050^{***}
Standard error	(0.040)	(0.039)	(0.018)	(0.011)
Population	No	Yes	No	No
Demographics	No	No	Yes	No
Socioeconomic	No	No	No	Yes
State effects	Yes	Yes	Yes	Yes
Observations	562	562	562	562
Clusters	8	8	8	8

State-clustered standard errors; normal-reference tests. $^{***}p < 0.01$. Only eight clusters; interpret significance cautiously.

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County estimates: columns 5–8

	(5)	(6)	(7)	(8)
Soybean intensity	-0.042	0.082^{***}	0.029^{*}	0.051^{***}
Standard error	(0.026)	(0.015)	(0.017)	(0.017)
Population	Yes	Yes	No	Yes
Demographics	Yes	No	Yes	Yes
Socioeconomic	No	Yes	Yes	Yes
State effects	Yes	Yes	Yes	Yes
Observations	562	562	562	562
Clusters	8	8	8	8

State-clustered standard errors; normal-reference tests. $^{*}p < 0.10$, $^{***}p < 0.01$. Only eight clusters.

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Why 562 counties do not create 562 independent shocks

- Counties within one state can share policy, administration, history, and economic shocks.
- State-clustered covariance allows residual dependence within states.
- The sample has eight state-level clusters, and soybean variation occurs within only five of them.
- Adding state fixed effects removes state intercepts; it does not remove all within-state residual correlation.

The reported normal-reference significance levels are therefore conditional on an inference procedure with few clusters. A large county count does not, by itself, resolve that limitation.

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The reported components of the sign reversal

Column (8), fully controlled soybean coefficient	0.051
Omitted income contribution	-0.064
Omitted poverty contribution	-0.017
Omitted unemployment contribution	-0.012
Column (5), without these controls	-0.042

$$0.051 - 0.064 - 0.017 - 0.012 = -0.042.$$

Conditional on the base controls, soybean-intensive counties have comparatively favorable income, poverty, and unemployment profiles. Their omission contributes negatively to the soybean coefficient.

▸ Decomposition check

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Economic improvement and conditional SNAP receipt

- Favorable general economic conditions can lower SNAP receipt.
- Holding those measured conditions fixed, the full specification associates historical soybean intensity with higher receipt.
- The residual conditional association could reflect other dimensions of need, eligibility, access, or historical institutions.
- The findings motivate the provisioning question but do not isolate entitlement erosion as its mechanism.

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Why more controls do not establish a direct causal effect

- Current income and employment may themselves reflect earlier agricultural transformation.
- Conditioning can remove part of a historical association or introduce selection through other common causes.
- County averages do not trace the households who left farming.
- With eight clusters, conventional significance classifications are particularly fragile.
- The exercise remains a geographically limited, conditional reduced-form comparison.

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Interpretation, Scope, and Conclusions

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The main findings together

- The two-sector calibration summarizes state differences in agricultural labor absorption conditional on productivity.
- Higher historical relative-margin indices are associated with lower calibrated subsistence.
- Historically tenant-oriented composition is associated with higher subsistence and different productivity–labor slopes.
- Historical soybean intensity predicts lower subsistence; the significant baseline IV relationship connects soybean-predicted subsistence negatively to SNAP.
- Tenure attenuates the first stage; suitability mainly reduces precision. The IV sign remains negative throughout.
- The county relationship changes with socioeconomic conditioning.

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What primitive accumulation contributes to the reading

- It asks how producers' alternatives and claims changed alongside measured efficiency.
- It connects productive expansion to the redistribution of control over land and other resources.
- It treats dependence on wages, markets, and public transfers as an institutional transformation to explain.
- The observed outcomes support an inductive interpretation when read alongside historical evidence.
- The regressions alone do not identify the complete process or its welfare balance.

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No presumption of an equitable agrarian past

- Plantation and tenancy relations could sustain livelihoods through unequal property and authority.
- Their withdrawal may matter without making their restoration desirable.
- Ownership could strengthen a claim while leaving an operator exposed to debt and market dependence.
- Marx and Engels distinguished critiques of capitalist relations from restoration of earlier exploitation.

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An interpretation without a single welfare verdict

- Structural transformation can raise incomes and reduce poverty.
- It can also weaken particular producers' claims to agricultural livelihoods.
- The empirical question concerns how these outcomes arose and whose command over subsistence changed.
- Neither persistence in agriculture nor exit alone establishes security, autonomy, or welfare improvement.

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The main inferential boundaries

Calibration	$\widehat{\alpha}$ is at the positivity boundary; $\widehat{a}_s$ depends on the model and its input data.
Farm dynamics	Stationary shape summaries do not identify annual rates or absolute margins. The common-cost mapping is conditional.
Tenure	PCA summarizes operator composition, not complete entitlement relations.
State IV	Relevance does not establish exclusion; the unweighted bootstrap interval includes zero.
County regressions	Few clusters, limited coverage, and contemporaneous controls constrain interpretation.
Historical interpretation	Several analyses share data and cannot be counted as independent tests of an entire causal narrative.

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A thin description with substantive content

The three slices bring different relationships into view:

- **Productivity and entitlement:** agricultural labor absorption and historically constituted access.
- **Profit and subsistence:** the distribution of operating control and conditional surplus claims.
- **Cash crops and SNAP:** commercialization and state-mediated food provisioning.

Together they ask how agricultural transformation redistributed the capacity to produce, appropriate, and secure the means of subsistence: an account that is thin, but not "thin."

▸ Check the connections ▸ References

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Appendix: Worked Checks and References

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Motivation: Productivity Accounting

- Worked accounting

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An accounting example with fixed prices

Worked example

	Initial situation	More output, same inputs
Revenue V	100	110
Measured costs C	80	80
Revenue–cost ratio H	1.25	1.375
Surplus above measured costs	20	30
Surplus share $1 - C/V$	20%	27.27%

- With one output and unchanged input quantities and prices, output and TFP rise by 10%.
- The accounts alone do not reveal who receives the extra surplus of 10.
- Redistributing the initial 20 without changing quantities would not itself raise TFP.

Hypothetical amounts; not reported empirical estimates.

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Slice 1: Allocation and Calibration

- Worked allocation
- Worked recovery

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An illustrative allocation calculation

For illustration only, let $\alpha = 0.01$, $\bar{a} = 0.02$, and $A_a = 0.4$.

$$\ell_a = 0.01 + 0.99\frac{0.02}{0.4} = 0.0595.$$

Doubling productivity to 0.8 gives

$$\ell_a = 0.01 + 0.99\frac{0.02}{0.8} = 0.03475.$$

- The agricultural labor share falls from 5.95% to 3.475%.
- The fixed subsistence parameter is unchanged in this calculation.
- A lower observed labor share could also reflect a different recovered $\bar{a}$.

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Recovering subsistence in a simple example

Worked example

Suppose a state's early-year labor share is 0.06, productivity is 0.4, and the candidate α is 0.01.

$$\bar{a}(0.01) = 0.4\frac{0.06 - 0.01}{0.99} = 0.020202\ldots$$

Substituting the recovered value back into the same observation gives

$$0.01 + 0.99\frac{0.020202\ldots}{0.4} = 0.06.$$

- A one-year inversion matches that recovery observation exactly, conditional on α.
- A two-year average generally need not match both years exactly.
- The six-date fit helps select the common α.

Hypothetical inputs; the estimated common parameter is not changed.

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Slice 2: Farm Dynamics and the Relative Index

- Markup example
- Count and land shares
- Recursion check
- Numerical distribution
- Ordering check
- Index example
- Invariance check
- CRRA comparison
- Coefficient calculation

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Markup and margin: a numerical distinction

A pricing example, not an empirical estimate

Let marginal cost be 80 and price be 100. Then

$$\Lambda = \frac{100}{80} = 1.25, \qquad \pi = \frac{100 - 80}{100} = 0.20.$$

Object	Calculation	Meaning
Markup above cost	$\Lambda - 1 = 25\%$	Relative to cost
Gross operating margin	$\pi = 20\%$	Relative to revenue

At revenue of 100, production labor costs 80 and gross operating profit is 20. Expansion costs and the model's omitted financial or rental expenses are not deducted in this margin.

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Counting farms and counting land: a small example

Hypothetical holdings: one acre, one acre, and eight acres

There are three farms operating ten acres. Normalize farm masses by ten units:

$$\mu_1 = \frac{2}{10}, \qquad \mu_8 = \frac{1}{10}, \qquad \sum_n n\mu_n = 1, \qquad F = \sum_n \mu_n = 0.3.$$

Size class	Share of farms	Share of land
One-acre farms	$2/3$	$2/10$
Eight-acre farms	$1/3$	$8/10$

Mean acreage is $1/F = 10/3$. A size class can contain few farms but most of the land. This accounting example need not be a stationary model distribution.

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Worked check: the lower distribution

Starting from $a_0 = 0$ and $a_n = pa_{n-1} + e/x$ for $n \le m$,

$$
\begin{aligned}
a_1 &= e/x, \\
a_2 &= (e/x)(1+p), \\
a_3 &= (e/x)(1+p+p^2), \\
a_n &= (e/x)\frac{1-p^n}{1-p}.
\end{aligned}
$$

Using $e/x = (1-p)/m$ gives $a_n = (1-p^n)/m$. Finally, $\mu_n = a_n/n$.

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A small numerical example of the distribution

Illustration only: let $m = 2$ and $p = 1/2$.

n	$a_n = n\mu_n$	μ_n	source
1	$1/4$	$1/4$	$(1-p)/m$
2	$3/8$	$3/16$	$(1-p^2)/m$
3	$3/16$	$1/16$	$a_2 p$
4	$3/32$	$3/128$	$a_2 p^2$

- These μ_n are masses. Divide them by F to obtain farm shares.
- The tail continues beyond size four.
- These illustrative values are not calibrated parameters for U.S. agriculture.

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Worked check: a larger p

Hold m and b fixed and compare two fitted environments.

- Larger p means incumbent acquisitions are closer to total displacement.
- The entrant-acquired share of displaced units, $1 - p$, is smaller.
- The stationary unit-weighted size distribution shifts upward.
- Mean model acreage rises and inverse p falls.
- Under the common-cost mapping, the conditional margin ranking falls.

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What an index value of two would mean

Illustrative normalization, not fitted reference moments

Suppose the reference mean of $h = 1/p$ is 1.001 and its standard deviation is 0.0005. For an observation with $h = 1.002$,

$$\mathcal{M} = \frac{1.002 - 1.001}{0.0005} = 2.$$

- This is two reference standard deviations above the reference mean.
- It does not mean a two-percent markup or a two-percentage-point margin.
- Small variation in h can produce large standardized values because the unit of measurement is its own standard deviation.

Under the common affine margin mapping, the same standardized value applies to the conditional margin; its unknown economic scale still cannot be recovered.

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Worked check: index invariance under affine margins

Illustration only. Suppose fitted inverse ratios are h_i. Two feasible common mappings might take the form

$$\pi_i^{(1)} = a_1 + b_1 h_i, \qquad \pi_i^{(2)} = a_2 + b_2 h_i, \qquad b_1, b_2 > 0.$$

For either mapping and the same reference set,

$$\frac{\pi_i^{(j)} - \bar{\pi}_{\mathcal{R}}^{(j)}}{\sigma_{\pi^{(j)},\mathcal{R}}} = \frac{h_i - \bar{h}_{\mathcal{R}}}{\sigma_{h,\mathcal{R}}} = \mathcal{M}_i.$$

The two mappings can imply different economic margin levels despite identical standardized scores.

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Worked check: what changes with CRRA utility?

For utility curvature γ, the Euler equation becomes

$$\frac{\dot{C}_s}{C_s} = \frac{r_s - \rho}{\gamma}.$$

On balanced growth,

$$r_s - g_s = \rho + (\gamma - 1)g_s.$$

If g_s differs across states and $\gamma \neq 1$, growth-adjusted discounting need not be common. The margin mapping then requires a new derivation and additional restrictions.

Teaching comparison only; the paper maintains log utility.

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Interpreting a coefficient in index units

For the full-sample, region-adjusted 1940 estimate,

$$\widehat{\Delta \log \bar{\tilde{a}}_s} = -0.628\, \Delta \mathcal{M}_s.$$

A one-unit difference in $\mathcal{M}$ is one 1940 reference standard deviation of the conditional margin.

$$\Delta \mathcal{M} = 1 \quad \Longrightarrow \quad 100\big(e^{-0.628} - 1\big) \approx -46.6\%.$$

This transforms the fitted log-index contrast into a proportional contrast. It is not a one-percentage-point change in observed profits or an estimated causal welfare effect.

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Slice 3: SNAP Associations and Conditioning

- Log-log calculation
- Interaction calculation
- Decomposition check

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Interpreting the log-log coefficient

For the unweighted IV estimate $\widehat{\beta} = -0.195$:

- A 1% increase in the instrument-predicted subsistence index corresponds approximately to a 0.195% decrease in the SNAP rate.
- For a 10% index increase, the exact fitted proportional contrast is

$$100\left[(1.10)^{-0.195} - 1\right] \approx -1.84\%.$$

This is a percent change in the participation rate, not a 1.84-percentage-point change. Causal interpretation additionally requires the maintained IV assumptions.

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An illustrative interaction calculation

Using the rounded estimates, at soybean share 0.01 the fitted slope is

$$-0.230 + 2.622(0.01) = -0.20378.$$

At share 0.05 it is

$$-0.230 + 2.622(0.05) = -0.0989.$$

- These are arithmetic examples, not estimates at documented representative sample values.
- The standard error of a conditional slope uses the covariance of $\widehat{\beta}_1$ and $\widehat{\beta}_3$.
- Separate coefficient stars do not establish significance at every share value.

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The sign reversal as a projection identity

After partialling out state effects, population, and demographics, write

$$\widetilde{y} = \beta_F \widetilde{x} + \widetilde{W}\gamma + \varepsilon, \qquad \widetilde{W}_j = \pi_j \widetilde{x} + v_j.$$

The coefficient omitting the socioeconomic variables satisfies

$$\beta_R = \beta_F + \sum_j \pi_j \gamma_j.$$

- π_j describes how a socioeconomic variable covaries with historical soybean intensity conditional on the base controls.
- γ_j describes its partial relationship with SNAP in the full regression.
- Their products account algebraically for the coefficient change on the same sample.

Same-sample OLS identity; not a causal mediation result.

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Worked check: connecting the three slices

Slice 1 supplies the outcome used in Slice 2:

$$\widehat{\bar{a}}_s = \frac{1}{|T_{early}|} \sum_{t \in T_{early}} A_{a,s,t} \frac{\ell_{a,s,t}^{obs} - \widehat{\alpha}}{1 - \widehat{\alpha}}, \qquad y_s = \log \widehat{\bar{a}}_s.$$

Slice 2 constructs a separate explanatory variable:

$$\mathcal{M}_{s,\tau} = \frac{\widehat{p}_{s,\tau}^{-1} - \bar{h}_{1940}}{\sigma_{h,1940}}, \qquad y_s = c + \beta \mathcal{M}_{s,\tau} + \text{region effects} + \varepsilon_s.$$

Slice 3 changes the outcome to log SNAP participation. Its OLS and IV specifications use the previously calibrated y_s; the IV first stage uses historical soybean intensity, not $\mathcal{M}$.

The modules share a calibrated quantity. They do not estimate one joint causal chain.

◂ Calibration ◂ Margin association ◂ SNAP association

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Discussion questions

1. Which assumption allows profit per unit to be independent of quality?
2. Why does m disappear from the incumbent expansion condition but remain in the size distribution?
3. Why is p^{-1} sufficient for a conditional ranking but insufficient for markup levels?
4. What information would convert the relative half-life into years?
5. Why does a positive empirical growth coefficient not estimate $g = x \log \Lambda$?

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Questions for an empirical extension

- Transition data on entry, exit, and transfers could discipline the calendar-time scale.
- Observed revenues, costs, and transaction terms could assess the pricing and margin interpretation.
- Household histories could distinguish loss of productive claims from voluntary compensated exit.
- Land-quality information could improve the mapping from acreage to productive units.

These are possible extensions. They are not additional analyses reported in the supplied paper.

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Prices and unit profits	Lemma 4.1	App. C.2
Incumbent policy and resources	Proposition 4.1	App. C.3–C.5
Stationary size distribution	Proposition 4.2	App. C.6
Ordering by model mean size	Lemma C.3	App. C.6
Relative adjustment speed	Proposition C.1	App. C.7
Conservation and affine mapping	Proposition 4.3	Section 4.6 / App. C.5, C.8
Relative margins and invariance	Proposition 4.4	App. C.8
Equilibrium growth ordering	Proposition 4.5	App. C.9
Historical fitting procedure	Calibration equations	App. C.10

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TFP and farm income	Section 2.2 / Appendix A	4–7 / 68–69
Structural model and calibration	Sections 3.1–3.4 / Appendix B	15–23 / 69–71
Historical tenure and PCA	Sections 3.5–3.7	23–29
Farm dynamics	Section 4 / Appendix C	29–45 / 71–88
Data and empirical strategy	Sections 5.1–5.8	46–51
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- The source manuscript
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- Representation, measurement, and selection
- Structural transformation and innovation models
- Empirical methods and public provisioning
- Historical institutions and agrarian change
- Agricultural accounts and data documentation

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